

Gear Manufacturing Process on Milling Machine

MODULE-1



Bahir Dar General Flexible Machinery and Foundry Factory



Done by: - Solomon Fisiha

Ethiopia, Bahir Dar

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Bahir Dar General Flexible Machinery and Foundry Factory**

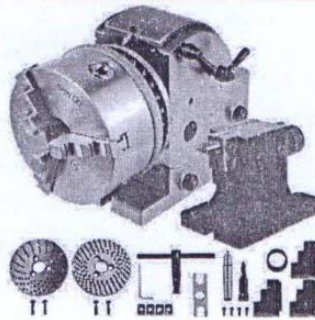
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Dividing head (Index)



Dividing head or index: - it a device for a milling machine attachment.

Use and application of dividing head

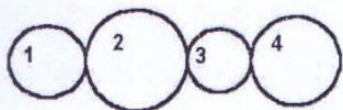
- ✦ To hold and support the work piece during the milling operation.
- ✦ To provide the circumference of the circle in to any number of equal parts.
- ✦ To provide a rotary motion to the work piece simultaneously with its longitudinal travel.

Parts of dividing head

- | | |
|------------------|-------------------------|
| 1). Index plate | 6). Worm shaft |
| 2). Index crank: | 7). Worm wheel |
| 3). Sector arm: | 8). Spindle |
| 4). Index pin: | 9). Foot stock |
| 5). Chuck: | 10). Direct index plate |

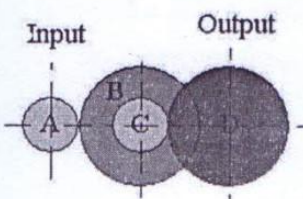
Gear train: The combination of different train of gears to transfer power.

Examples of Simple gear train given below



$$\text{Gear Ratio} = \frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} * \frac{Z_3}{Z_2} * \frac{Z_4}{Z_3} = \frac{Z_4}{Z_1}$$

Example of compound gear train given below



$$\text{Gear Ratio} = \frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{B}{A} * \frac{D}{C}$$

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There are different types of indexing

1). Direct indexing (plane indexing)

- ✦ Direct indexing is accomplished by an additional index plate fastened to the index head spindle. A stationary plunger in the index head fits the holes in this index plate.
- ✦ The direct indexing is useful and practical for achieving 2,3,4,6,8,12 and 24 division by a means of direct index plate which is supplied with the dividing head. i.e. $\frac{24}{N}$, where N is number of division required.
- ✦ Any number of divisions which is a factor of 24 can be indexed quickly and conveniently by the direct indexing method.
- ✦ The spindle is moved by hand.
- ✦ The worm shaft and the worm wheel are disengaged (disconnecting).
- ✦ It is performed by gradual ring on the spindle nose on the dividing head.
- ✦ The spindle must be locked by the spindle lock nut.

Ex1:- Find out the index movement required to mill a hexagonal bolt by direct indexing.

Solutions: number of hole to be moved $= \frac{24}{6} = 4$. After machining one side of the bolt the index plate will have to be moved by 4 holes for 5 number times to machine the remaining face of the bolt.

2). Simple indexing (indirect indexing).

- ✦ It is an indexing that cannot be obtained from direct indexing.
- ✦ The worm shaft and the worm wheel are engaged (connecting).
- ✦ The ratio of indexing crank in the spindle is 40:1
- ✦ When the spindle makes one full rotation the indexing crank makes 40 turn. Therefore the formula will be:
 $T = \frac{40}{N}$, where T = crank movement, N = number of teeth to be cut, 40 = number of teeth on the worm wheel gear.

Ex1:- A gear wheel with 17 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66)

Solution: - $T = \frac{40}{Z} = \frac{40}{17}$ = Since 17 plate number are not supplied with standard plate number listed the above. Multiply both the nominator and denominator by 2 to get the standard set of plate. i.e. $2 \frac{6 \times 2}{17 \times 2} = 2 \frac{12}{34}$, thus the index crank should be 2 complete turn and moved 12 holes on a 34 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Ex2:- A gear wheel with 52 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66)

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Solution: - $T = \frac{40}{Z} = \frac{40}{52}$ Since 52 plate number are not supplied with standard plate number listed the above. This means that less than one full turn is required for each division. $40/52$ must be reduced to its lowest term, divide both the nominator and denominator by 4. The lowest term which is **10/13**.

Therefore, Take the denominator of the lowest term 13, and determine into which of the available hole circles it can be evenly divided. In this case, 13 can be divided into a 39-hole circle exactly 3 times. Use this result 3 as a multiplier to generate the proportional fraction required. $T = \frac{10 \cdot 3}{13 \cdot 3} = \frac{30}{39}$. Therefore, 30 holes on a 39-hole circle is the correct indexing for 52 divisions. When counting holes, start with the first hole ahead of the index pin.

Ex3:- A gear wheel with 25 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

plate1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: - $T = \frac{40}{Z} = \frac{40}{25}$ Since 25 plate number are not supplied with standard plate number listed the above. Take the lowest term $1\frac{3}{5}$ multiply both the nominator and denominator by 3 to get the standard set of plate. i.e $1\frac{3 \cdot 3}{5 \cdot 3} = 1\frac{9}{15}$, thus the index crank should be 1 complete turn and moved 9 holes on a 15 hole circle is the correct indexing for 25 for division. When counting holes, start with the first hole ahead of the index pin.

3). Differential indexing

- We use differential indexing when it is not possible to make an equidistant division with simple indexing. In this case the required rotation accomplished by connecting the index plate to the spindle by gear train. The gear train between the dividing head spindle and the index plate provides the correct ratio of movement between the spindle and the index plate.
- During differential indexing the index plate locking pin should be taken out to make the plate free to rotate. The bevel gear shaft and the dividing head spindle are connected by a gear train.
- In differential indexing the proper ratio of gearing is found in the following manner:

✦ **Step1:** select a number of divisions that can be indexed by simple indexing. It can be either greater or smaller than the required number. Determine the movement of index crank using the formula: turn of index crank $T = \frac{40}{A}$, where, A= the approximate number of division.

✦ **Step2:** determine the gear ratio from the formula: gear ratio $= \frac{\text{driver gear}}{\text{driven gear}} = \frac{\text{gear on spindle head}}{\text{gear on bevel gear shaft}} = (A-N) \cdot \frac{40}{A}$, where, A= the approximate number of division from step1, N= the required number of divisions.

If (A-N) is positive, the index plate must be rotated in the same direction as the crank. If (A-N) is negative, the index plate must be rotated in the opposite direction to the crank.

To achieve these conditions, the number of idle gears used depends on the following factors.

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- a). if the gear train is simple and (A-N) is positive, only one idler gear is used .
 - b). if the gear train is compound and (A-N) is positive, no idler gear is used.
 - c). if the gear train is simple and (A-N) is negative, two idler gear is used.
 - d). if the gear train is compound and (A-N) is negative, only one idler gear is used.
- ✦ **Step3:** select the suitable gears from the standard change gears with which the index head is equipped. **The ratio of gearing will determine whether a simple or a compound train of gear is used.**

Ex1:- Calculate the indexing and change gear required for 57 division. Assume that standard change gear are 24, 24, 28, 32, 40, 44, 48, 56, 64, 72, 86, 100 and the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: - Assume that A=56, Crank movement $T = \frac{40}{A} = \frac{40}{56}$ = Since 56 plate number are not supplied with standard plate number listed the above. Divide the nominator and denominator by 8. i.e. $= \frac{5}{7}$, multiply both by 3 to get the standard set of plate. Therefore $T = \frac{5 \times 3}{7 \times 3} = \frac{15}{21}$, Therefore, 15 holes on a 21-hole circle is the correct indexing for 57 divisions.

Change gears $i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{\text{gear on spindle head}}{\text{gear on bevel gear shaft}} = (A-N) \times \frac{40}{A} = (56-57) \times \frac{40}{56} = \frac{40}{56}$ in these case use simple gear train because 40 and 56 are available on the standard set of gear.

Ex2:- Calculate the indexing and change gear required for 67 division. Assume that standard change gear are 25, 25, 30, 35, 40, 50, 55, 60, 70, 80, 90, 100 and the available index plate hole circle are as follow: plete1: 24, 25, 28, 30, 34, Plate2: 37, 38, 39, 41, 42, 43 Plate3: 46, 47, 49, 51, 53, and plate4: 54, 57, 58, 59, 62, 66

Solution: - Assume that A=70, Crank movement $T = \frac{40}{A} = \frac{4}{7}$ = Since 7 plate number are not supplied with standard plate number listed the above. Multiply the nominator and denominator by 4 to get the standard set of plate. i.e. $T = \frac{4 \times 4}{7 \times 4} = \frac{16}{28}$, Therefore, 16 holes on a 28-hole circle is the correct indexing for 67 divisions.

Change gears $i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{\text{gear on spindle head}}{\text{gear on bevel gear shaft}} = (A-N) \times \frac{40}{A} = (70-67) \times \frac{40}{70} = 3 \times \frac{4}{7} = \frac{12}{7}$ since 12 and 7 tooth gear are not supplied with standard set of gear listed the above. It is necessary to multiply the 12/7 ratio by a number that will suit the standard gear available. To get the standard set of gear multiply the nominator and the denominator by 5.

Change gears $i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{12 \times 5}{7 \times 5} = \frac{60 \text{ on the spindle}}{35 \text{ on the worm}}$

Ex3:- Calculate the indexing and change gear required for 271 division. Assume that standard change gear are 24, 24, 28, 32, 40, 44, 48, 56, 64, 72, 86, 100 and the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

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Solution: - Assume that $A=280$, Crank movement $T = \frac{40}{A} = \frac{40}{280} = \frac{4}{28}$. Since 28 plate number are not supplied with standard plate number listed the above. Divide the nominator and denominator by 4 i.e. $T = \frac{1}{7}$, multiply both by 3 to get the standard set of plate. Therefore $T = \frac{1 \times 3}{7 \times 3} = \frac{3}{21}$. Therefore, 3 holes on a 21-hole circle is the correct indexing for 271 divisions.

$$\text{Change gears } i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{\text{gear on spindle head}}{\text{gear on bevel gear shaft}} = (A-N) \times \frac{40}{A} = (280-271) \times \frac{40}{280} = 9 \times \frac{40}{280} = \frac{9}{7}$$

The fraction $\frac{9}{7}$ when raised to obtain the numbers equivalent to the available set of gears. Multiply both the nominators and the denominators by 8 i.e. $\frac{9}{7} \times \frac{8}{8} = \frac{72 \text{ on the spindle}}{56 \text{ on the worm}}$.

Ex4:- Calculate the indexing and change gear required for 319 division. Assume that standard change gear are 24, 24, 28, 32, 40, 44, 48, 56, 64, 72, 86, 100 and the available index plate hole circle are as follow: plate1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: - Assume that $A=290$, Crank movement $T = \frac{40}{A} = \frac{40}{290} = \frac{4 \text{ holes}}{29 \text{ circle}}$. Therefore, 4 holes on a 29-hole circle is the correct indexing for 319 divisions.

$$\text{Change gears } i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{\text{gear on spindle head}}{\text{gear on bevel gear shaft}} = (A-N) \times \frac{40}{A} = (290-319) \times \frac{40}{290} = -29 \times \frac{40}{290} =$$

$-29 \times \frac{4}{29} = -\frac{4}{1}$ in these case there is no a number to multiplied and to get the standard set of gear. Therefore simple gear train is not used instead compound gearing is used i.e. $\frac{4}{1} \times \frac{3}{3} = \frac{12}{3} = \frac{4 \times 3}{1 \times 3}$. It is now necessary to multiply both the nominator and the denominator of each fraction by the same number in order to bring the ratio in to the range of the gears available on the standard set of gear. Therefore,

$$\text{Change gears } i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{4}{3} \times \frac{16}{16} = \frac{64}{48}, \text{ 64 and 48 are available on the above listed standard set of gear.}$$

Change gears $i = \frac{\text{driver gear}}{\text{driven gear}} = \frac{3}{1} \times \frac{24}{24} = \frac{72}{24}$, 72 and 24 are available on the above listed standard set of gear. Finally, change gears = $\frac{\text{the product of driver gears}}{\text{the product of driven gears}} = \frac{72 \times 64}{24 \times 48}$, the negative sign indicate the index plate must be rotated in the opposite direction to the crank. The gear train is compound, in these case only using one idler gear.

The required gear train is $\frac{a}{b} \times \frac{c}{d}$, where, a = the number of teeth of the gear on the spindle shaft.

b = the first gear on the stub.

c = the second gear on the stub.

d = the number of teeth of the gear on the shaft.

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4). Angular indexing

- ✚ The worm gear drive with a ratio of 40:1 between the index arm and the dividing head spindle permits indexing in degrees in connection with the index plate. It is obvious that as 40 revolutions of the crank give one complete revolution of the spindle, one turn of the crank will give $\frac{360}{40} = 9^\circ$
- ✚ The formula for this index will be $T = \frac{1 \text{ degree}}{9}$

Ex1:- Index 1° use the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20
Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: $T = \frac{1 \text{ degree}}{9} = \frac{1}{9}$ Since 9 plate number are not supplied with standard plate number listed the above. Multiply both by 2 to get the standard set of plate. $\frac{1 \times 2}{9 \times 2} = \frac{2}{18}$, thus the index crank should be moved 2 holes on a 18 hole circle for each division giving 1° .

Ex2:- Index 43° use the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20
Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: $T = \frac{1 \text{ degree}}{9} = \frac{43}{9} = 4\frac{7}{9}$, since 9 plate number are not supplied with standard plate number. $4\frac{7 \times 2}{9 \times 2} = 4\frac{14}{18}$, thus the index crank should be 4 complete turn and moved 14 holes on a 18 hole circle for each division giving 43° .

Ex3:- Index 72° use the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20
Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: $T = \frac{1 \text{ degree}}{9} = \frac{72}{9} = 8 \text{ turns.}$

Simple indexing can also be used to index fractional parts of a degree. Using B and S dividing head, it is possible to index $1/2, 2/3, 1/3$

Ex4:- Index $72\frac{1}{2}$ degree use the available index plate hole circle are as follow: plete1: 15, 16, 17, 18, 19, 20
Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: $T = \frac{1 \text{ degree}}{9} = \frac{72\frac{1}{2}}{9} = 8 \text{ turn} \frac{1}{2}$, Since 9 plate number are not supplied with standard plate number. Multiply both by 2 to get the standard set of plate. $T = 8 \frac{1}{2} \times \frac{2}{2} = 8 \frac{1}{18}$. Thus the index crank should be 8 complete turn and moved 1 holes on a 18 hole circle for each division giving $72\frac{1}{2}$ degree.

Ex5:- Index $5\frac{1}{2}$ degree, use the available index plate hole circle are as follow:

Plete1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49



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Solution: $T = \frac{1 \text{ degree}}{9} = \frac{5 \frac{1}{2}}{9}$ = Since 9 plate number are not supplied with standard plate number.

Multiply both by 2 to get the standard set of plate. $T = \frac{5 \frac{1}{2}}{9} * \frac{2}{2} = \frac{11}{18}$, thus the index crank should be moved 11 holes on a 18 hole circle for each division giving $5 \frac{1}{2}$ degree.

Ex6:- Index $6 \frac{2}{3}$ degree, use the available index plate hole circle are as follow:

Plate1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: $T = \frac{1 \text{ degree}}{9} = \frac{6 \frac{2}{3}}{9} = \frac{(6*3)+2}{9}$, Since 9 plate number are not supplied with standard plate

number. Multiply both by 3 to get the standard set of plate. $T = \frac{(6*3)+2}{9} * \frac{3}{3} = \frac{20}{27}$, thus the index crank should be moved 20 holes on a 27 hole circle for each division giving $6 \frac{2}{3}$ degree.

54- hole circle on the standard plate provided by the Cincinnati dividing head attachment permits the index of $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{3}$, and $\frac{1}{6}$



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Spur Milling

Spur gear: A straight tooth gear, is a type of cylindrical gear with teeth that are cutting parallel to gear's axis. The two parallel and co-planar shafts connected by the gears. These gears are called *spur gears* and the arrangement is known as *spur gearing*.

Spur gears can be manufactured from a variety of materials, each offering different properties and suitability for specific applications. Some common materials used for spur gears include: steel, cast iron, brass, bronze, plastics, and aluminum.

Spur gears are subjected to various types of loads during operation, and understanding these loads is crucial for designing and selecting the appropriate gears for specific applications. Some of the key loads that spur gears experience include: tangential load, radial load, axial load, over load or shock load and dynamic load.

The general formula for spur gears will be explained as follows:

1). $D_o = m(z+2)$ where, D_o =Outer diameter of the work piece, z = number of teeth to be cut,

m =module. But $m = \frac{D_o - p}{z+2 \pi}$ where, p = pitch of the work piece.

2). $p = \pi * m$ where, p = pitch of the work piece, m =module.

3). $d = m * z$ where, d = pitch circle diameter, m =module, z = number of teeth.

4). $d_f = D_o - 2h_f$ where, d_f =root circle diameter, h_f =dedendum, D_o =Outer diameter of the work piece.

5). $h_a = m$ where, h_a = addendum, m =module.

6). $h_f = m + c$ where, h_f = dedendum, m =module, c = tooth tip clearance.

7). $c = \frac{1}{6} * m$ where, c = tooth tip clearance, m =module.

8). $h = \text{addendum} + \text{dedendum} = h_a + h_f = m + m + c = 2m + c$ where, h = total depth, m =module, c = tooth tip clearance.

9). $a = \frac{D_{o1} + D_{o2}}{2}$ where, a =center distance between the two axis, D_o = Outer diameter of the work.

10). $t = 0.5 * p$ where, t =thickness of the teeth, p = pitch of the work piece.

11). $T = \frac{40}{N}$ where, T = index crank movement, N = number of teeth to be cut, 40= number of teeth on the worm wheel gear.



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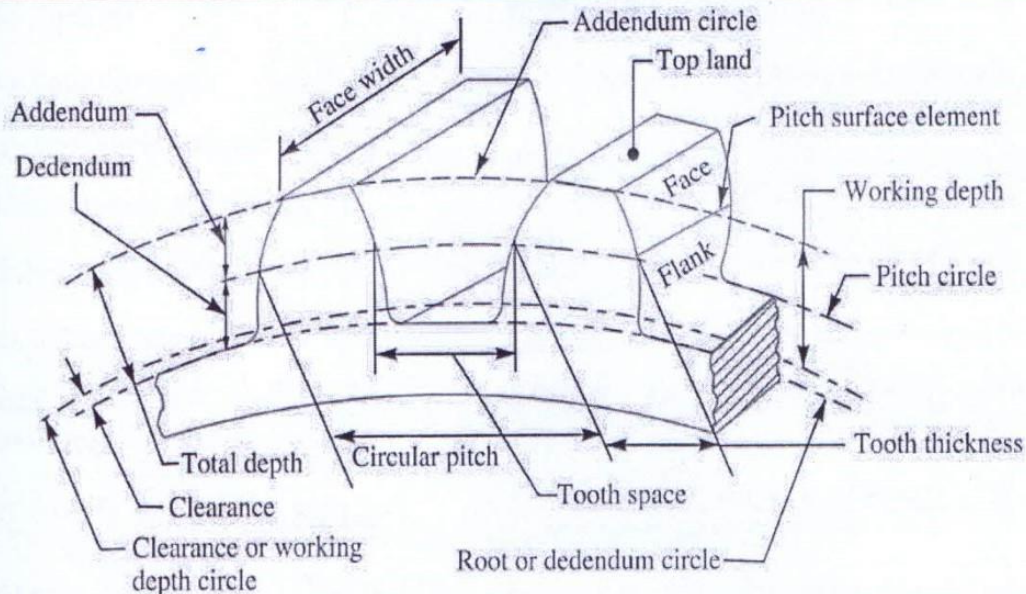


Fig: nomenclature of gear

Ex1:- A spur gear having number of teeth is 40 and outside diameter 252mm to be milled. Calculate?

- | | |
|----------------------|------------------------|
| a). Module of cutter | F). crank movement |
| b). Total depth | G). pitch |
| c). Pitch diameter | H). tip clearance |
| d). Dedendum | I). thickness of teeth |
| e). Addendum | |

Given data: $D_o = 252\text{mm}$, $Z = 40$,

Solution: a). $D_o = m(z+2)$, $m = \frac{D_o}{z+2} = \frac{252}{40+2} = \frac{252}{42} = 6$, b). $h = 2m + c$, but $c = \frac{1}{6} * m = \frac{1}{6} * 6 = 1$, therefore, $h = (2*6) + 1 = 13\text{mm}$ c). $d = m * z = 6 * 40 = 240\text{mm}$ d). $h_f = m + c = 6 + 1 = 7\text{mm}$ e). $h_a = m = 6$

f). $T = \frac{40}{N} = \frac{40}{40} = 1 \text{ full turn}$ g). $p = \pi * m = \pi * 6 = 18.84\text{mm}$ h). $c = \frac{1}{6} * m = \frac{1}{6} * 6 = 1$

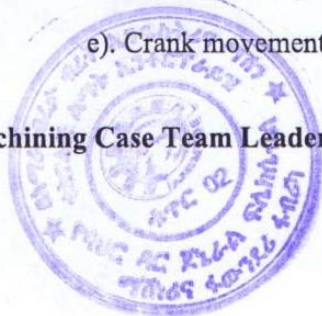
i). $t = 0.5 * p = 0.5 * 18.84 = 9.42\text{mm}$

Ex2:- A spur gear having number of teeth is 27 and the module of the cutter is 2.5. Use a standard set plate number given below: Assume that standard change gear are 25, 25, 30, 35, 40, 50, 55, 60, 70, 80, 90, 100 and the available index plate hole circle are as follow: plate1: 24, 25, 28, 30, 34, Plate2: 37, 38, 39, 41, 42, 43 Plate3: 46, 47, 49, 51, 53, and plate4: 54, 57, 58, 59, 62, 66. Calculate:

- | | |
|--------------------------------|------------------------|
| a). Outside diameter (D_o) | e). Crank movement (T) |
|--------------------------------|------------------------|

Prepared By: Solomon Fisiha (Machining Case Team Leader)

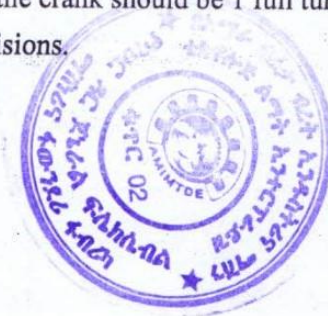
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b). Pitch (p)	f). Thickness of the gear (t)
c). Total depth (h)	g). Change gear ratio? If we use differential indexing
d). Root circle diameter (d_f)	

Solution: a). $D_o = m(z+2) = 2.5(27+2) = \underline{72.5\text{mm}}$ b). $p = \pi \cdot m = \pi \cdot 2.5 = \underline{7.85\text{mm}}$
c). $h = 2m + c = 2m + \frac{1}{6}m = (2 \cdot 2.5) + \frac{1}{6} \cdot 2.5 = \underline{5.42\text{mm}}$ d). $d_f = D_o - 2h_f$, but $h_f = m + c = 2.5 + (\frac{1}{6} \cdot 2.5) = \underline{2.92}$. Therefore, $d_f = D_o - 2h_f = 72.5 - 2 \cdot 2.92 = \underline{66.66\text{mm}}$
e). $T = \frac{40}{N} - \frac{40}{27} = 1 \frac{13 \cdot 2}{27 \cdot 2} = 1 \frac{26}{54}$ (by simple indexing).

Solution: a). $p = \pi * m = \pi * 3 = \underline{9.42\text{mm}}$,
 b). $h = 2m + c = 2m + \frac{1}{6} * m = (2 * 3) + \frac{1}{6} * 3 = \underline{6.5\text{mm}}$,
 c). $t = 0.5 * p = 0.5 * 9.42 = \underline{4.71\text{mm}}$
 d). $h_f = m + c = 3 + (\frac{1}{6} * 3) = \underline{3.5\text{mm}}$,
 e). $d_f = D_o - 2h_f = 78 - 2 * 3.5 = \underline{71\text{mm}}$,
 f). $d = m * z = 3 * 24 = \underline{72\text{mm}}$
 g). $T = \frac{40}{24} = 1\frac{16}{24}$, Therefore, the crank should be 1 full turn and 16 holes on a 24-hole circle is the correct indexing for 24 divisions.
 h). $c = \frac{1}{6} * m = \frac{1}{6} * 3 = \underline{0.5\text{mm}}$



Helical Milling



A helical gear has teeth in form of helix around the gear. Two such gears may be used to connect two parallel shafts in place of spur gears. The helixes may be right handed on one gear and left handed on the other. The pitch surfaces are cylindrical as in spur gearing, but the teeth instead of being parallel to the axis, wind around the cylinders helically like screw threads. The teeth of helical gears with parallel axis have line contact, as in spur gearing. This provides gradual engagement and continuous contact of the engaging teeth. Hence helical gears give smooth drive with a high efficiency of transmission.

In order to cut either a metric or an inch helix, any two of the following must be known:-

- a) **The lead of the helix (L):-** it is the longitudinal distance the helix advances axially in one complete revolution of the work piece.
- b) **The angle of the helix (β):-** is performed by the intersection of the helix with the axis of the work piece.
- c) **The Pitch circle diameter (d) or circumference of the pitch circle diameter of work piece.**

➤ Diagrammatically given bellow

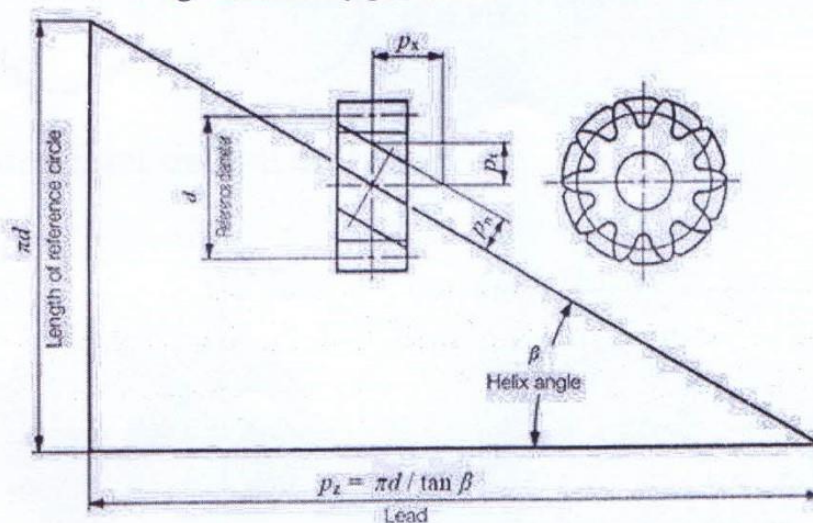


Fig: The relationship between circumference, lead and helix angle.

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In comparing two different helices, it will be noticed that the greater the angle with the center line, the shorter will be the lead. If the diameter is increased but the helix angle remains the same, the greater will be the lead. Thus it is evident that the lead of the helix varies with: the diameter of the work piece and the helix angle.

Cutting a helix: to cut a helix on a cylinder, the following steps are necessary:-

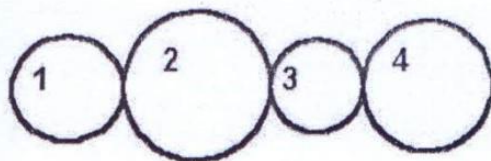
- 1) Swing the table in proper direction to the angle of the helix.
- 2) The work must rotate one turn while the table travels lengthwise the distance equal to the lead. This is achieved by installing the proper change gears between the worm shaft on the dividing head, and on the milling machine lead screw.

Determine the direction to swing the milling machine table:-

- If the helix slopes down to the right, it is a right hand helix. To cut right helix, moving the right end of the table towards the column or by moving it in a counterclockwise direction. If you use a compound gear train, use zero or even number idler gears for right hand helix. But if you use a simple gear train use one or add number idler gears for right hand helix.
- If the helix slopes down to the left, it is left hand helix. To cut left helix, moving the left end of the table towards the column or by moving it in a clockwise direction. If you use a compound gear train, use one or odd number idler gears for left hand helix. But if you use a simple gear train use zero or even number idler gears for left hand helix.

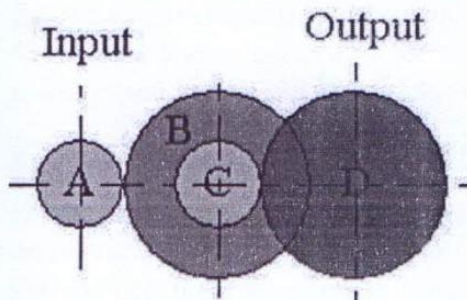
Gear train: - The combination of different gears to transfer power.

Examples of Simple gear train given below



$$\text{Gear Ratio} = \frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} * \frac{Z_3}{Z_2} * \frac{Z_4}{Z_3} = \frac{Z_4}{Z_1}$$

Example of compound gear train given below



$$\text{Gear Ratio} = \frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{B}{A} * \frac{D}{C}$$



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The general formula for helical gears will be explained as follows:

1). $L = \frac{\pi d}{\tan \beta}$ where, L= lead of helix, d= pitch circle diameter of the work piece, β = helix angle (the milling machine swiveled).

Ex1:- To what angle must a milling machine table be swiveled to cut a helix having a lead of 450mm on a work piece of 40mm of a pitch circle diameter?

Solution: Tangent of helix = $\tan \beta = \frac{\pi d}{L} = \frac{\pi * 40}{450} = \frac{125.66}{450} = 0.27925$ therefore, $\beta = \tan^{-1}(0.27925) = 15.6^\circ$

2). $D_o = d + 2m$ where, D_o = Outer diameter of the work piece, d= pitch circle diameter, m= module

3). $d = m_t * z$ where, d= pitch circle diameter, m_t = tangential module, z= number of tooth

But $m_t = \frac{m_n}{\cos \beta}$ where, m_n = Normal module β = helix angle, Therefore, $d = \frac{m_n}{\cos \beta} * z$

4). $P_t = \pi * m_t$ where, P_t = transvers (tangential) pitch, m_t = tangential module.

5). $P_n = \pi * m_n$ where, P_n = normal pitch, m_n = normal module. But $m_n = m_t * \cos \beta$.

Therefore, $P_n = \pi * m_t * \cos \beta = P_t * \cos \beta$.

6). $t = \frac{P_n}{2} = 0.5 * P_n$, where, t=normal tooth thickness, P_n = normal pitch.

7). $h_a = m_n$ where, h_a = addendum, m_n =normal module.

8). $h_f = 1.25 * m_n$ where, h_f = dedendum, m_n =normal module.

9). $h = \text{addendum} + \text{dedendum} = h_a + h_f = m_n + 1.25 * m_n = 2.25 * m_n$, where, h= full depth, m_n = normal module.

10). $d_f = d - 2.5 * m_n$ where, d_f =root circle diameter, d= pitch circle diameter, m_n =normal module.

11). $c = \frac{1}{6} * m_n$ where, c= tooth tip clearance, m_n = normal module.

12). Change gears = $\frac{\text{product of driven gears}}{\text{product of driver gears}} = \frac{\text{lead of helix to be cut (mm)}}{\text{lead of machine (mm)}} = \frac{L}{p * 40}$ where, L=lead of helix, p=pitch of lead screw on the milling.

Ex2:- Calculate the change gears required to cut a helix having a lead of 450mm on a work piece of 40mm in diameter. using a standard set of gears (25,30,35,40,50,55,60,70,80,90,100).The milling machine lead screw has a pitch of 5mm.

Solution: - Case 1. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p * 40} = \frac{450}{5 * 40} = \frac{450}{200} = \frac{45}{20}$, Since 45 and 20 tooth gear are not supplied with standard set of gear. It is necessary to multiply both the nominator and the denominator by a number that will suit the standard gear available. Therefore to get the standard set of gear multiply the nominator and the denominator by 2.



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Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{45 \times 2}{20 \times 2} \times \frac{90}{40} = 2.25$, driven gear (attached on the dividing head worm shaft) = 90 and driver gear (attached on milling machine lead screw) = 40. In this case the gear arrangement is simple gear train.

Solution: - Case 2. Using a standard set of gears (24, 24, 28, 32, 38, 40, 44, 48, 56, 64, 72, 86, 100). Change gears = $\frac{\text{product of driven gears}}{\text{product of driver gears}}$ From the universal precision dividing head table for spiral milling for the lead of helix having 450mm and for the 5mm pitch of lead screw you can get the number of tooth of the compound gear. i.e. $Z_1=24, Z_2=72, Z_3=64, Z_4=48$, the gear ratio of case 1 and case 2 must be same. $\frac{Z_2}{Z_1} \times \frac{Z_4}{Z_3} = \frac{72}{24} \times \frac{48}{64} = 2.25$, in this case the gear arrangement is compound gear train.

$Z_4=48$ (Attached on the dividing head), $Z_3=64$ (where it is attached to Z_4 gear on bracket holders).

$Z_2=72$ (where it is attached to Z_1 gear on bracket holders), $Z_1=24$ (Attached on the milling machine lead screw).

NB: Z_3 and Z_2 fix with the same shaft on the compound gearing system. The preceding order is not absolutely necessary; the two driven gears may be interchanged and/ or the two driver gears may be used interchangeably, provided a driver is not interchanged with a driven gears.

Ex3:- Calculate the change gears required to cut a helix having a lead of 500mm on a work piece using a standard set of gears(24,24,28,32,38,40,44,48,56,64,72,86,100). The milling machine lead screw has a pitch of 5mm.

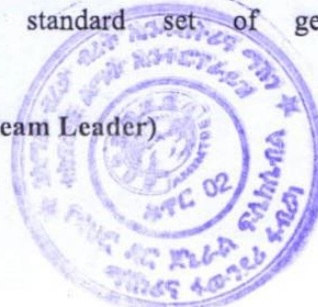
Solution: - Case 1. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p \times 40} = \frac{500}{5 \times 40} = \frac{500}{200} = \frac{5}{2}$, Since 5 and 2 tooth gear are not supplied with standard set of gear. It is necessary to multiply the 5/2 ratio by a number that will suit the standard gear available. Therefore to get the standard set of gear multiply the nominator and the denominator by 20.

Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{5 \times 20}{2 \times 20} \times \frac{100}{40} = 2.5$, driven gear (attached on the dividing head worm shaft=100) and driver gear (attached on milling machine lead screw=40).

In this case the gear arrangement is simple gear train.

Solution: - Case 2. Change gears = $\frac{\text{product of driven gears}}{\text{product of driver gears}}$ From the universal precision dividing head table for spiral milling for the lead of helix having 500mm and for the 5mm pitch of lead screw you can get the number of tooth of the compound gear. i.e. $Z_1=24, Z_2=72, Z_3=48, Z_4=40$, the gear ratio of case 1 and case 2 is the same. $\frac{Z_2}{Z_1} \times \frac{Z_4}{Z_3} = \frac{72}{24} \times \frac{40}{48} = 2.5$, in this case the gear arrangement is compound gear train.

Ex4:- Calculate the change gears required to cut 14 number of helical tooth having a lead of 1500mm on a work piece, having a helix angle of 6 degree, having a module of 3.5, and a milling machine lead screw has a pitch of 5mm. using a standard set of gears given below. (25,30,35,40,50,55,60,70,80,90,100).



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Case 1. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p \cdot 40} = \frac{1500}{5 \cdot 40} = \frac{1500}{200} = \frac{15}{2}$, Since 15 and 2 tooth gear are not supplied with standard set of gear. In these case there is no a number to multiplied and to get the standard set of gear. Therefore simple gear train is not used instead compound gearing is used i.e. $\frac{15}{2} = \frac{5 \cdot 3}{2 \cdot 1}$. Take the ratio $\frac{5}{2}$ and $\frac{3}{1}$ independently for compound gearing. *Finally it does not change the value of the fraction.*

Change gears $i = \frac{5}{2}$, It is necessary to multiply the $\frac{5}{2}$ ratio by a number that will suit the standard gear available. $\frac{5 \cdot 20}{2 \cdot 20} = \frac{100}{40}$, 100 and 40 are available on the above listed standard set of gear.

Change gears $i = \frac{3}{1}$, It is necessary to multiply the $\frac{3}{1}$ ratio by a number that will suit the standard gear available. $\frac{3 \cdot 30}{1 \cdot 30} = \frac{90}{30}$, 90 and 30 are available on the above listed standard set of gear.

Finally, change gears = $\frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} \cdot \frac{Z_4}{Z_3} = \frac{100 \cdot 90}{40 \cdot 30} = \frac{7.5}{1}$

Solution: Case2. Change gear Ratio = $\frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{40 \cdot p \cdot \sin \beta}{\pi \cdot m \cdot n \cdot z} = \frac{40 \cdot 5 \cdot \sin 6}{\pi \cdot 3.5 \cdot 14} = \frac{20.9056}{153.938} = 0.13580$,

Now from the table you can get the number of tooth of the compound gear. $\frac{Z_2}{Z_1} \cdot \frac{Z_4}{Z_3} = \frac{55}{30} \cdot \frac{100}{25} = \frac{7.33}{1}$

$Z_1=30, Z_2=55, Z_3=25, Z_4=100$, the gear ratio of case 1 and case two approximately same

Ex5:- Calculate the change gears required to cut a helix having a lead of 1680mm on a work piece using a standard set of gears : (24,24,28,32,38,40,44,48,56,64,72,86,100). The milling machine lead screw has a pitch of 6mm.

Solution: -Case1. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p \cdot 40} = \frac{1680}{6 \cdot 40} = \frac{1680}{240} = \frac{42}{6}$, Since 42 and 6 tooth gear are not supplied with standard set of gear. In these case there is no a number to multiplied and to get the standard set of gear. Therefore simple gear train is not used instead compound gearing is used.

Therefore, $\frac{42}{6} = \frac{14 \cdot 3}{6 \cdot 1}$. It is now necessary to multiply both the nominator and the denominator of each fraction by the same number in order to bring the ratio in to the range of the gears available on the standard set of gear. Take the ratio $\frac{14}{6}$ and $\frac{3}{1}$ independently for compound gearing. *Finally it does not change the value of the fraction.*

Change gears $i = \frac{14}{6}$, It is necessary to multiply the $\frac{14}{6}$ ratio by a number that will suit the standard gear available. $\frac{14 \cdot 4}{6 \cdot 4} = \frac{56}{24}$, 56 and 24 are available on the above listed standard set of gear.

Change gears $i = \frac{3}{1}$, It is necessary to multiply the $\frac{3}{1}$ ratio by a number that will suit the standard gear available. $\frac{3 \cdot 24}{1 \cdot 24} = \frac{72}{24}$, 72 and 24 are available on the above listed standard set of gear.

Finally, change gears = $\frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} \cdot \frac{Z_4}{Z_3} = \frac{56 \cdot 72}{24 \cdot 24} = 7$

Prepared By: Solomon Fisiha (Machining Case Team Leader)



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NB: - Z2 and Z3 fix with the same shaft on the compound gearing system. The preceding order is not absolutely necessary; the two driven gear may be interchanged and/or the two driver gears may be used interchangeably, provided a driver is not interchanged with a driven gears.

Case2. From the universal precision dividing head table for spiral milling for the lead of helix having 1680mm and for the 6mm pitch of lead screw you can get the number of tooth of the compound gear. i.e. $Z_1=40, Z_2=84, Z_3=24, Z_4=80$, the gear ratio of case 1 and case 2 is the same. $\frac{Z_2}{Z_1} * \frac{Z_4}{Z_3} = \frac{84}{40} * \frac{80}{24} = 7$

Ex6:- Calculate the change gears required to cut 13 number of helical tooth having a lead of 1100mm, outside diameter of 130mm on a work piece and having a helix angle of 18 degree, and a milling machine lead screw has a pitch of 5mm. using a standard set of gears: (25,25,30,35,40,50,55,60,70,80,90,100). And also calculate the necessary data?

Solution: -Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p * 40} = \frac{1100}{5 * 40} = \frac{1100}{200} = \frac{11}{2}$, Since 11 and 2 tooth gear are not supplied with standard set of gear. In these case there is no a number to multiplied and to get the standard set of gear. Therefore simple gear train is not used instead compound gearing is used. Multiply both the nominator and denominator by 5. Therefore, $\frac{11}{2} * \frac{5}{5}$. Take the ratio $\frac{11}{5}$ and $\frac{5}{2}$ independently for compound gearing. **Finally it does not change the value of the fraction.**

Change gears $i = \frac{11}{5}$, It is necessary to multiply the 11/5 ratio by a number that will suit the standard gear available. $\frac{11 * 5}{5 * 5} = \frac{55}{25}$, 55 and 25 are available on the above listed standard set of gear.

Change gears $i = \frac{5}{2}$, It is necessary to multiply the 5/2 ratio by a number that will suit the standard gear available. $\frac{5 * 20}{2 * 20} = \frac{100}{40}$, 100 and 40 are available on the above listed standard set of gear.

Finally, change gears = $\frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} * \frac{Z_4}{Z_3} = \frac{55 * 100}{25 * 40} = 5.5$

Z_1 is attached on the milling machine lead screw, Z_2 and Z_3 is attached on the gear bracket holder and Z_4 attached on the dividing head.

NB: - Z2 and Z3 fix with the same shaft on the compound gearing system. The preceding order is not absolutely necessary; the two driven gear may be interchanged and/or the two driver gears may be used interchangeably, provided a driver is not interchanged with a driven gears.

$D_o = d + 2m$, but $d = \frac{m * z}{\cos \beta}$, therefore $D_o = \frac{m * z}{\cos \beta} + 2m$, $130 = \frac{m * 13}{\cos 18} + 2m$, $130 = 1.0515 * 13m + 2m$,

$130 = m * (1.0515 * 13 + 2)$, $130 = 15.67m$, therefore, $m = \frac{130}{15.67} = 8.29$. $d = D_o - 2m$, $d = 130 - 2 * 8.29 = 113.4mm$

$h = 2.25 * m = 2.25 * 8.29 = 18.65mm$



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Ex7:- Calculate the change gears required to cut 13 number of helical tooth having a lead of 1100mm on a work piece, having a helix angle of 18 degree, and a milling machine lead screw has a pitch of 5mm. using a standard set of gears: (24,24,28,32,38,40,44,48,56,64,72,86,100).

Solution: -Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{L}{p \cdot 40} = \frac{1100}{5 \cdot 40} = \frac{1100}{200} = \frac{11}{2}$, Since 11 and 2 tooth gear are not supplied with standard set of gear. In these case there is no a number to multiplied and to get the standard set of gear. Therefore simple gear train is not used instead compound gearing is used. Multiply both the nominator and denominator by 6. Therefore, $\frac{11}{2} \cdot \frac{6}{6}$. Take the ratio $\frac{11}{6}$ and $\frac{6}{2}$ independently for compound gearing. **Finally it does not change the value of the fraction.**

Change gears $i = \frac{11}{6}$, It is necessary to multiply the 11/6 ratio by a number that will suit the standard gear available. $\frac{11 \cdot 4}{6 \cdot 4} = \frac{44}{24}$, 44 and 24 are available on the above listed standard set of gear.

Change gears $i = \frac{6}{2}$, It is necessary to multiply the 6/2 ratio by a number that will suit the standard gear available. $\frac{6 \cdot 12}{2 \cdot 12} = \frac{72}{24}$, 72 and 24 are available on the above listed standard set of gear.

Finally, change gears = $\frac{\text{the product of driven gears}}{\text{the product of driver gears}} = \frac{Z_2}{Z_1} \cdot \frac{Z_4}{Z_3} = \frac{44 \cdot 72}{24 \cdot 24} = 5.5$. **The gear ratio must be same as the above example.**

To cut a helix on an inch milling machine: - The following necessary data is used to get the desired change gears.

A Standard milling machine lead screw has 4 threads/inch. The lead screw, as it revolves one turn, would revolve the dividing head spindle 1/40 of a revolution. in order for the dividing head spindle to revolve one turn, it would be necessary for the lead screw to revolve 40 times. Thus the table would travel $40 \cdot \frac{1}{4 \text{ in}}$. Therefor the lead of the machine is **10in**. In calculating the change gears required to cut any lead the following formula may be used.

$$13). \text{ Change gears} = \frac{\text{driven gears} \cdot \text{lead of helix to be cut (in)}}{\text{driver gears} \cdot \text{lead of machine (in)}} = \frac{L}{10 \text{ in}}$$

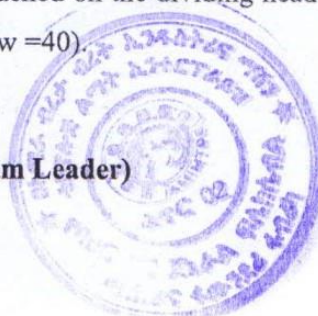
Ex8:- Calculate the change gears required to cut a helix having a lead of 25in on a work piece using a standard set of gears(24,24,28,32,38,40,44,48,56,64,72,86,100). The milling machine lead screw has a lead of 10in.

Solution: - Change gears = $\frac{\text{driven gears} \cdot \text{lead of helix to be cut (in)}}{\text{driver gears} \cdot \text{lead of machine (in)}} = \frac{25}{10}$. Since 25 and 10 tooth gear are not supplied with standard set of gear. It is necessary to multiply the 25/10 ratio by a number that will suit the standard gear available. To get the standard set of gear multiply the nominator and the denominator by 4.

Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{25 \cdot 4}{10 \cdot 4} = \frac{100}{40}$, driven gear (attached on the dividing head worm shaft = 100) and driver gear (attached on milling machine lead screw = 40).

In this case the gear arrangement is **simple gear train**.

Prepared By: Solomon Fisiha (Machining Case Team Leader)



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Ex9:- Calculate the change gears required to cut a helix having a lead of 27in on a work piece using a standard set of gears (24,24,28,32,38,40,44,48,56,64,72,86,100). The milling machine lead screw has a lead of 10in.

Solution: - Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{\text{lead of helix to be cut (in)}}{\text{lead of machine (in)}} = \frac{27}{10}$, since there are no gears in the set which are multiples of both 27 and 10, it is impossible to use simple gear train. **Compound gearing** must therefore be used. And it becomes necessary to factor the fraction $\frac{27}{10}$ as follows.

Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{27}{10} = \frac{3 \cdot 9}{2 \cdot 5}$ (driven), Take the ratio $\frac{3}{2}$ and $\frac{9}{5}$ independently for compound gearing. It is now necessary to multiply both the nominator and the denominator of each fraction by the same number in order to bring the ratio in to the range of the gears available on the standard set of gear. **Finally it does not change the value of the fraction.**

Therefore. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{3 \cdot 16}{2 \cdot 16} = \frac{48}{32}$, 48 and 32 are available on the standard set of gear.

Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{9 \cdot 8}{5 \cdot 8} = \frac{72}{40}$, 72 and 40 are available on the set of the standard gear. Finally.

Change gears = $\frac{\text{product of driven gears}}{\text{product of driver gears}} = \frac{48 \cdot 72}{32 \cdot 40}$, the driven gear are 48 and 72, the driver gear are 32 and 40.

Z₄=72= Attached on the dividing head.

Z₃=40= where it is attached to Z₄ gear on bracket holders.

Z₂=48= where it is attached to Z₁ gear on bracket holders.

Z₁=32= Attached on the milling machine lead screw.

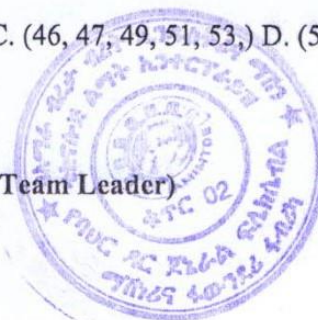
NB: Z₃ and Z₂ fix with the same shaft on the **compound gearing system**. The preceding order is not absolutely necessary; the two driven gears may be interchanged and/ or the two driver gears may be used interchangeably, provided a driver is not interchanged with a driven gears.

14). Change gear ratio = $\frac{40 \cdot p \cdot \sin \beta}{\pi \cdot m_n \cdot z}$ where, p=pitch of lead screw, β= helix angle, m_n=normal module, z= no of tooth.

15). $T = \frac{40}{Z}$ Where, T= index crank movement, Z= number of teeth to be cut, 40= number of teeth on the worm wheel gear.

Ex10: - A gear wheel with 17 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66)



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Solution: - $T = \frac{40}{Z} = \frac{40}{17}$ = Since 17 plate number are not supplied with standard plate number. $2 \frac{6 \times 2}{17 \times 2} = 2 \frac{12}{34}$, thus the index crank should be 2 complete turn and moved 12 holes on a 34 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Ex11: - A gear wheel with 52 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66)

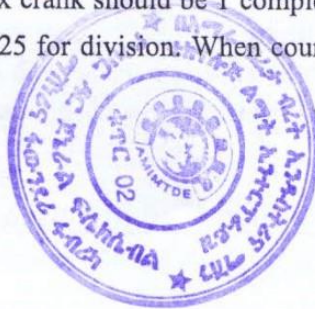
Solution: - $T = \frac{40}{Z} = \frac{40}{52}$ = Since 52 plate number are not supplied with standard plate number. This means that less than one full turn is required for each division. $40/52$ must be reduced to its lowest term, divide both the nominator and denominator by 4. The lowest term which is **10/13**.

Therefore, Take the denominator of the lowest term 13, and determine into which of the available hole circles it can be evenly divided. In this case, 13 can be divided into a 39-hole circle exactly 3 times. Use this result 3 as a multiplier to generate the proportional fraction required. $T = \frac{10 \times 3}{13 \times 3} = \frac{30}{39}$, Therefore, 30 holes on a 39-hole circle is the correct indexing for 52 divisions. When counting holes, start with the first hole ahead of the index pin.

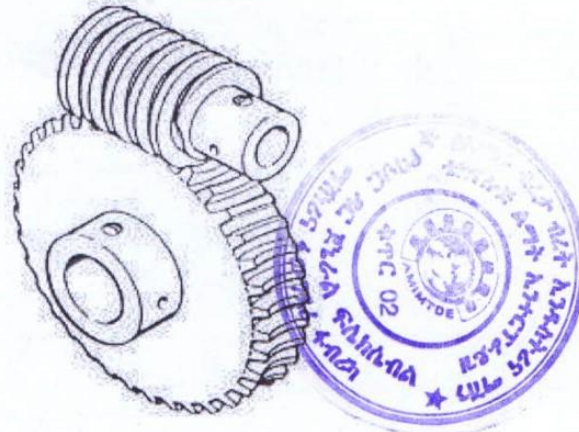
Ex12: - A gear wheel with 25 teeth is to be milled find the number of crank movement and the number of holes. Use a standard set plate number given below:

plate1: 15, 16, 17, 18, 19, 20 Plate2: 21, 23, 27, 29, 31, 33 Plate3: 37, 39, 41, 43, 47, 49

Solution: - $T = \frac{40}{Z} = \frac{40}{25}$ = Since 25 plate number are not supplied with standard plate number listed the above. $1 \frac{3}{5} = 1 \frac{3 \times 3}{5 \times 3} = 1 \frac{9}{15}$, thus the index crank should be 1 complete turn and moved 9 holes on a 15 hole circle is the correct indexing for 25 for division. When counting holes, start with the first hole ahead of the index pin.



Worm shaft



Worm shaft: it is a cylindrical rod with a helical thread wrapped around it. Its primary function is to transmit rotational motion between non-intersecting perpendicular axes. Provide high gear reduction ratio and used machinery, automotive steering systems, dividing head, conveyor system. Some commonly used materials for worm shaft are: steel alloy, bronze alloy, and aluminum alloy. And worm shaft in worm gear can experience various types of load such as: radial load, thrust load, torsional load, and dynamic load.

It is necessary to consider the relationship between the number of starts on the worm and the efficiency to decide the suitability of worm gear drive for a particular application.

Two guidelines are as follows:

1. Single-threaded worm gives large speed reduction, however, the efficiency is low.
2. Multi-threaded worm gives high efficiency, however, the speed reduction is low.

In a worm gear drive, power is always transmitted from worm to worm wheel. Power cannot be transmitted from worm wheel to worm. This phenomenon is called self-locking. It is highly useful in many applications. Velocity ratio is determined by the number of teeth on worm gear and the number of starts on worm.

The general formula for worm shaft will be explained as follows:

- 1). $D_o = d + 2m$, where, D_o = Outside diameter of shaft, d = Pitch diameter of worm shaft, m = module.
- 2). $d = D_o - 2m$, where, D_o = Outside diameter of shaft, d = Pitch diameter of worm shaft, m = module.
- 3). $P = \pi \cdot m \cdot n$, where, P = Pitch and m = module, n = number of start.
- 4). $l = p \cdot n$ where, l = lead, p = pitch, n = number of start.

Lead. It is the linear distance through which a point on a thread moves ahead in one revolution of the worm. For single start threads, lead is equal to the axial pitch, but for multiple start threads, lead is equal to the product of axial pitch and number of starts. For a given pitch, the lead angle is controlled principally by two factors: (1) **the number of threads** and (2) **the pitch diameter of the worm**.

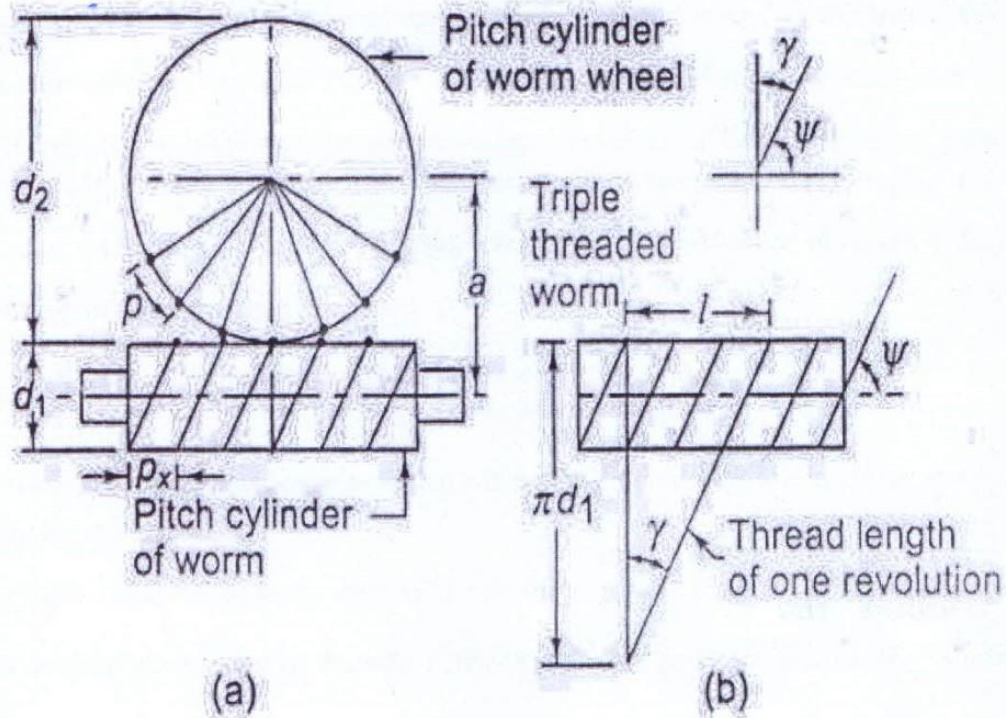


Fig: Relationship between lead and pitch circle diameter.

Lead angle. It is the angle between the tangent to the thread helix on the pitch cylinder and the plane normal to the axis of the worm. It is denoted by γ .

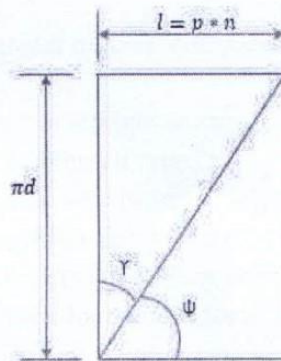
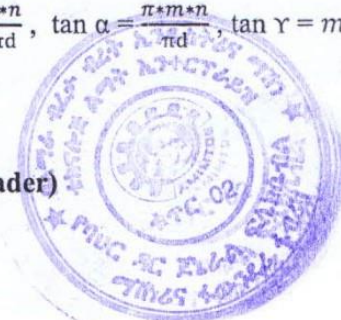


Fig. Development of a helix thread.

Where, πd = Pitch circumference of the worm, l = lead, d = pitch circle diameter of the worm shaft.

$$\tan \gamma = \frac{\text{lead of the worm}}{\text{pitch circumference of the worm}}, \quad \tan \gamma = \frac{l}{\pi d}, \quad \tan \gamma = \frac{p \cdot n}{\pi d}, \quad \tan \alpha = \frac{\pi \cdot m \cdot n}{\pi d}, \quad \tan \gamma = m \cdot \frac{n}{d}$$



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5). $\gamma = \tan^{-1}\left(m \cdot \frac{n}{d}\right)$, where, γ = Lead angle, where worm wheel on the milling Machine should be swiveled at this angle, n = number of start, d = pitch circle diameter of the worm shaft and p = Pitch

The lead angle (γ) may vary from 9° to 45° . Lead angle less than 9° results in rapid wear.

6). Helix angle (ψ) = It is the angle between the tangent to the thread helix on the pitch circle diameter and the axis of the worm. The worm helix angle is the complement of worm lead angle, i.e. $\gamma + \psi = 90^\circ$.

7). The cutter tip formula will be $W \leq 0.697 \cdot m$, where, m = module of cutter, w = width of the cutter tip.

8). The length of the worm shaft thread will be:

- ✓ $L = (13-16) \cdot m$ where, the number of start is (1-2).
- ✓ $L = (15-20) \cdot m$ where, the number of start is (3-4).

9). Indexing on the compound rest dimension will be $I = \frac{P}{N}$ where, I = indexing on the compound rest
 P = Pitch, N = number of start.

10). The depth of cut for the worm shaft will be $h = 2.2 \cdot m$ where, h is depth of cut, m = module.

11). The modular pitch found on the lathe machine table will be $M_p = \frac{P}{\pi}$ where, M_p = Modular pitch,
 P = Pitch.

N.B

- ✱ The Pitch of the worm wheel and shaft should be equal unless cannot mat each other.
- ✱ The direction of cut for worm wheel and worm shaft should be the same direction.
- ✱ Lead angle for worm shaft and wheel should be the same to mat each other.
- ✱ It is used for speed reduction.
- ✱ It is used to convert the horizontal motion to vertical motions.
- ✱ It is two types of worm shaft, thus:
 1. Cylindrical or straight worm.
 - ✓ It is common type.
 - ✓ It uses $14\frac{1}{2}^\circ$ pressure angle cutters. For single and double start and
 - ✓ For triple and quaderable start it uses 20° .
 2. Cone or double enveloping worm.
 - ✓ Is used for some extents. But its alignment is accuracy.

Ex1:- If the worm shaft number of start is 2 and the module cutter is 1.5 and also the diameter of worm shaft is 40mm in addition to the cut of direction is right. Now calculate the necessary data.

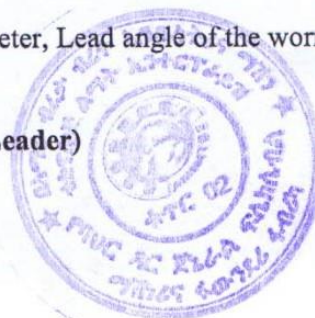
Given Data: Number of start (n) = 2, $D_o = 40\text{mm}$, $m = 1.5$, Cut of direction = Right

The necessary data to be calculated are as follows:

$d = D_o - 2m$, $d = 40 - (2 \cdot 1.5)$ $d = \underline{37\text{mm}}$ = pitch circle diameter, Lead angle of the worm shaft

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$\gamma = \tan^{-1}(m * \frac{n}{d})$, $\gamma = \tan^{-1}(1.5 * \frac{2}{37})$, $\gamma = 4.6^\circ$. In which, the worm wheel cutting on the milling machine table is swiveled. Cutting tip width will be calculated as $W = 0.697 * m = 0.697 * 1.5 = 1.05 \text{ mm}$

Length of the worm shaft will be calculated as $L = (13-16) * m$, $L = 14.5 * 1.5$, $L = 21.75 \text{ mm}$

The Pitch of the worm shaft will be calculated as $P = \pi * m * n = 3.14 * 1.5 * 2 = 9.42 \text{ mm}$

Indexing compound rest should be calculated as $I = P/n = 9.42/2 = 4.71 \text{ mm}$, Depth of cut can be calculated as $h = 2.2 * m = 2.2 * 1.5 = 3.3 \text{ mm}$, The Modular Pitch will be calculated as $M_p = P/\pi$

$$M_p = 9.42/3.14 = 3$$

Ex2:- if the worm shaft number of start is 3 having the module 2.5 and also the diameter of the worm shaft is 50mm in addition to the cut of direction is right. Now calculate the necessary data?

Given Data: Number of start (n) = 2 $D_o = 50 \text{ mm}$, $m = 2.5$, Cut of direction = Right

The necessary data to be calculated are as follows:

Solution: $P = \pi * m * n = \pi * 2.5 * 3 = 23.55 \text{ mm}$, $M_p = \frac{P}{\pi} = \frac{23.55}{\pi} = 7.5$, $I = \frac{P}{n} = \frac{23.55}{3} = 7.85$

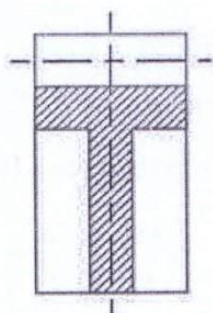
$W \leq 0.697 * m = 0.697 * 2.5 \leq 1.742$, the lead angle of the worm shaft. $\tan \gamma = m * \frac{n}{d}$, but $d = D_o - 2m$

$d = 50 - (2 * 2.5) = 45 \text{ mm}$, therefore, $\tan \gamma = 2.5 * \frac{3}{45} = 0.16666$, $\gamma = \tan^{-1} 0.16666 = 9.46^\circ$. In which, the worm wheel cutting on the milling machine table is swiveled. $h = 2.2 * m = 2.2 * 2.5 = 5.5 \text{ mm}$

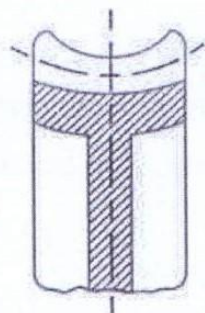
Worm wheel

Worm wheel: it is a gear that meshes with a worm shaft to provide rotational motion. It consists of toothed wheel.

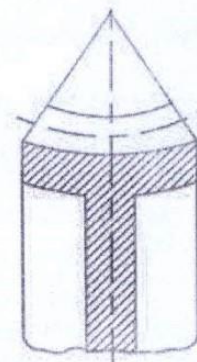
There is three types of worm wheel.



(a) Straight face.



(b) Hobbed straight face.



(c) Concave face.

a). Straight face worm gear. b). Hobbed Straight face worm gear. c). Concave face worm gear.



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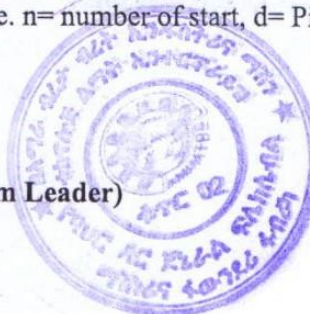
- ✦ Straight face worm gear.
 - It is like helical gear in which straight teeth are cut with a form cutter.
 - It has only point contact with worm threads. Therefore; it is used for light services.
- ✦ Hobbed Straight face worm gear.
 - It is used for light services.
 - Its teeth are cut with hobbing.
- ✦ Concave face worm gear.
 - It is standard acceptance type of worm gear.
 - It is used for heavy duty services and general industrial uses.

General consideration for cutting worm gears are as follows:

- ✦ If the lead angle is between 0° & 16° , the pressure angle for the cutter should be $14\frac{1}{2}^\circ$.
- ✦ If the lead angle is between 16° & 25° , the pressure angle will be 20° .
- ✦ If the lead angle is between 25° & 35° , the pressure angle will be 25° .
- ✦ If the lead angle is between 35° & 45° , the pressure angle will be 30° . It used for automotive application since it has high efficiency.

The general formula for worm wheel will be explained as follow:

- 1). $D \leq D_3 + 2m$, when the start number is one, D = Outside diameter of the worm wheel, D_3 = groove diameter and m = module.
- 2). $D \leq D_3 + 1.5m$, when the start number is two or three.
- 3). $D \leq D_3 + m$, when the start number is four.
- 4). $D_3 = m(Z+2)$, where, D_3 = groove diameter and m = module, z = number of teeth.
- 5). the thickness of the worm wheel will be calculated as:
 - ✦ $B \leq 0.75 \cdot D_o$, when the number of start should be ≤ 3 . Where, W = thickness of the wheel. D_o = Outside diameter of worm shaft.
 - ✦ $B \leq 0.67 \cdot D_o$, when the number of start should be four.
- 6). Center distance between shaft and wheel can be calculated as (the average of pitch circle of them)
 - ✦ $A = \frac{(D_o - 2m) + Z \cdot m}{2} = \frac{d + Z \cdot m}{2}$ Where, D_o = Outside diameter of the worm shaft, Z = number of teeth on worm wheel, m = module cutter. d = pitch diameter of the worm shaft
- 7). Lead angle of worm wheel can be calculated as : (Always use the pitch circle diameter of the worm).
 - ✦ $\tan \gamma = \frac{p}{\pi d}$, $\gamma = \tan^{-1} \left(m \cdot \frac{n}{d} \right)$, Or
 - ✦ $\gamma = \tan^{-1} \left(m \cdot n \right) / (D_o - 2m)$, Where, P = Pitch, γ = Lead angle, where worm wheel on the milling machine should be swiveled at this angle. n = number of start, d = Pitch circle for worm shaft, D_o = outside diameter of the worm shaft.



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Ex1:- If the number of teeth of the worm gear is 30 and the module cutter number is 1.5. Now calculate the necessary data of the worm wheel if the worm shaft number of start is 2, having the worm shaft pitch circle diameter is 37mm and having 40mm outside diameter of the worm shaft.

Given Data: $Z = 30$, $m = 1.5$, Number of start = 2, pitch circle diameter of the worm shaft = 37mm, outside diameter of the worm shaft = 40mm.

Solution: The necessary data will be calculated as follows:

When the start number is two or three use the formula $D \leq D_3 + 1.5m$ but $D_3 = m(Z+2) = 1.5(30+2)$, $D_3 = 48$ mm, $D \leq D_3 + 1.5m$, $D \leq 48 + 1.5 \times 1.5$, **$D \leq 50.25$ mm.**

Width of the worm wheel will be calculated as follows: number of start of the worm shaft is 2 and outer diameter of the worm shaft is 40mm. Use the formula: $B \leq 0.75 \times D_o$, $B \leq 0.75 \times 40$, **$B \leq 30.0$ mm.**

The center distance between the shaft and wheel will be calculated as follows:

$$A = \frac{D_o - 2m + Z \cdot m}{2}, \quad A = \frac{40 - 2 \times 1.5 + 30 \times 1.5}{2} = \mathbf{41 \text{ mm}}$$

The lead angle of the worm wheel will be calculated as follows:

$$\gamma = \tan^{-1} \left(m \cdot \frac{n}{d} \right) =, \quad \gamma = \tan^{-1} \left(1.5 \cdot \frac{2}{37} \right) = \mathbf{4.6^\circ} \text{ in which the milling machine table is swiveled.}$$

T = index crank movement, use the standard set of plate given below:

A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66).

Therefore, $T = \frac{40}{N} = \frac{40}{30} = 1 \frac{10}{30}$, thus the index crank should be 1 complete turn and moved 10 holes on a 30 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Ex2:- A triple threaded worm has teeth of 6 module and pitch circle diameter of 50mm. if the worm gear has 30 number of teeth. Calculate the lead angle and the center distance.

Solution: The lead angle of the worm shaft. $\tan \gamma = m \cdot \frac{n}{d} = 6 \cdot \frac{3}{50} = 0.36$, $\gamma = \tan^{-1} 0.36 = \mathbf{19.8^\circ}$

$$\text{Center distance} = \frac{D_o - 2m + Z \cdot m}{2} = \frac{d + Z \cdot m}{2}, \quad A = \frac{50 + 30 \cdot 6}{2} = \mathbf{115 \text{ mm}}$$



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Rack milling

A rack and pinion gear system consists of a linear gear, called a 'rack,' and a circular gear, called a 'pinion.' The rack is a straight or linear gear with teeth that mesh with the teeth of the pinion gear.

The pinion is small, a circular gear that engages with the teeth of the rack and converts rotational motion in to linear motion or vice versa. Making them suitable applications such as steering system in vehicles, CNC machines, industrial automations, and other machinery that requires precise linear motion control.

Some common material used for rack gear is steel, stainless steel, aluminum and plastics.

Rack gears are subjected to various loads and forces during operation, include: tangential load, radial load, axial load, over load, shock load and dynamic load.

The general formula for rack gears will be explained as follows:

- 1). $p = \pi * m$ where, p =pitch, m =module.
- 2). $s = \frac{\pi * m}{2}$ where, s =tooth thickness, m =module.
- 3). $c = 0.25 * m$ where, c =clearance tip m =module.
- 4). $h_a = m$ where, h_a = addendum m =module.
- 5). $h_f = 1.25 * m$ where, h_f = dedendum m =module.
- 6). $h = 2.25 * m$ where, h = full depth m =module.
- 7). $L = z * p$ where, L =length of the work piece, z =number of tooth to be cut, p =pitch.

Ex1:- Calculate the length of the work piece of the number of gear teeth is having 60 and pitch between the teeth is 2.5 and also calculate the module of cutter the depth of cut on the work piece.

Given data: $z=60$, $p=2.5$

Solution: length of the work piece = $z * p = 60 * 2.5 = \underline{150\text{mm}}$, module (m) = $\frac{p}{\pi} = \frac{2.5}{\pi} = \underline{0.796\text{mm}}$

Full depth (h) = $2.25 * m = 2.25 * 0.796 = \underline{1.791\text{mm}}$

Ex: Calculate the number of teeth if the length of the work piece is 120mm and the pitch between the gears is 3mm .Calculate:

- | | | |
|---------------------|-------------------|---------------------|
| a). Module | c). Depth | e). Tooth thickness |
| b). Number of teeth | d). Clearance tip | f). Dedendum |

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Given data: $L=120$, $p=3$,

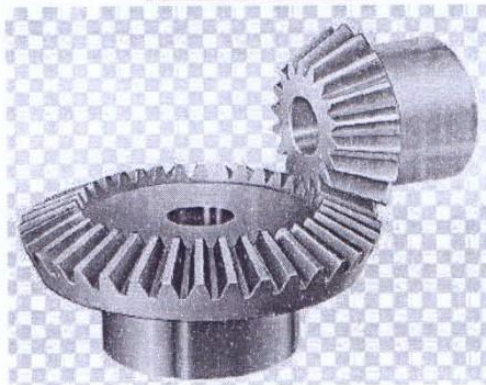
Solution: module $(m) = \frac{p}{\pi} = \frac{3}{\pi} = \underline{0.95\text{mm}}$, Number of teeth $(Z) = \frac{L}{p} = \frac{120}{3} = \underline{40}$,

Full depth $(h) = 2.25*m = 2.25*0.95 = \underline{2.148\text{mm}}$, Clearance tip $(c) = 0.25*m = 0.25*0.95 = \underline{0.2375\text{mm}}$,

Tooth thickness $(s) = \frac{\pi*m}{2} = \frac{\pi*0.95}{2} = \underline{1.4922\text{mm}}$, Dedendum $(h_f) = 1.25*m = 1.25*0.95 = \underline{1.1875\text{mm}}$



Bevel Milling



Bevel Gear: It is a type of gear that has cone-shaped teeth and is used to transmit power between two shafts that are not parallel to each other. The bevel gears are used for transmitting power at a constant velocity ratio between two shafts whose axis intersect at a certain angle. They are commonly used in applications such as differential gears in automobile, power tools, and industry machinery.

Bevel gear can be classified as: **Straight bevel gears, Spiral bevel gears and Hypoid bevel gears.**

Depending on the shape and orientation of their teeth: Mitre gears, Angular gears, crown gear and internal gears.

1. **Mitre gears.** When equal bevel gears (having equal teeth and equal pitch angles) connect two shafts whose axes intersect at right angle.
2. **Angular bevel gears.** When the bevel gears connect two shafts whose axes intersect at an angle other than a right angle, then they are known as **angular bevel gears.**
3. **Crown bevel gears.** When the bevel gears connect two shafts whose axes intersect at an angle greater than a right angle and one of the bevel gears has a pitch angle of 90°.
4. **Internal bevel gears.** When the teeth on the bevel gear are cut on the inside of the pitch cone.

Bevel gears are subjected to various types of loads, including: Axial load, radial load, bending load, shear load, torsional load.

Bevel gears are commonly made from a variety of materials, depending on the specific application and the required properties of the gear. Some of the materials used for bevel gears include: steel, cast iron, bronze, aluminum and plastics.

The general formula for Bevel Gear will be explained as follow:

1). $D = m(z + 2 \cos \theta) = mz + 2m \cos \theta$. But $mz = d$, therefore, $D = d + 2m \cos \theta$, where, D = outside diameter of the bevel gear, z = the number of gear tooth, m = module cutter, d = pitch diameter, θ = pitch cone angle.

2). $R = \frac{mz}{2 \sin \theta}$, Where, R = radius length of cone up to vertex (Pitch cone distance). From the pitch circle cone line. We can get the pitch cone radius. Therefore, $\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{\frac{d}{2}}{R} \Rightarrow R \sin \theta = \frac{d}{2}$. Therefore, $R = \frac{d}{2 \sin \theta}$, but, $d = m \cdot z$, then, $R = \frac{d}{2 \sin \theta} = \frac{m \cdot z}{2 \sin \theta}$, where m = module, z = number of teeth, d = pitch circle diameter.

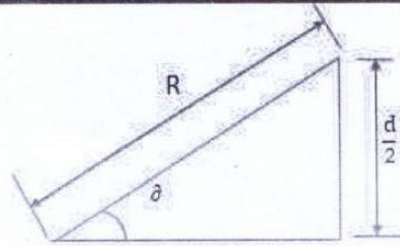


Fig: relationship between pitch cone radius and pitch circle diameter.

3). $\delta = \delta_f + \phi_f$, where, δ_f root angle at which the dividing head is swiveled. ϕ_f dedendum angle

The pitch cone angle is also calculated as follow: $\delta_1 = \tan^{-1} \left(\frac{\sin \Sigma}{\frac{z_2}{z_1} + \cos \Sigma} \right)$, and $\delta_2 = \tan^{-1} \left(\frac{\sin \Sigma}{\frac{z_1}{z_2} + \cos \Sigma} \right)$. Where

δ_1 = pinion pitch cone angle, z_1 = number of teeth on the pinion, z_2 = number of teeth on the gear and δ_2 = gear pitch cone angle and Σ = shaft angle. Generally, shaft angle $\Sigma = 90^\circ$ is most used. Other angles are sometimes used. Then, it is called "bevel gear in non-right angle drive". The 90° case is called "bevel gear in right angle drive". If the shaft angle is 90° the pitch cone angle should be calculated as follow $\delta_1 = \tan^{-1} \left(\frac{z_1}{z_2} \right)$, and $\delta_2 = \tan^{-1} \left(\frac{z_2}{z_1} \right)$.

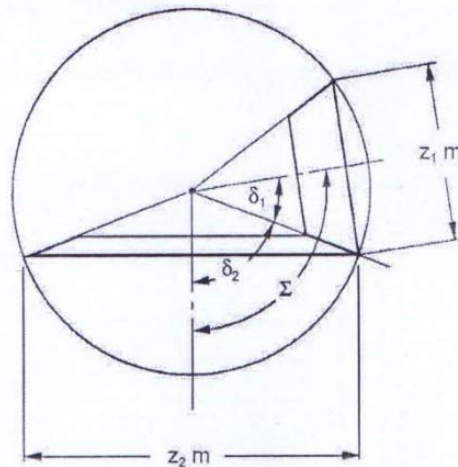


Fig: The relationship between shaft angle and pitch cone angle of the mating bevel gear.

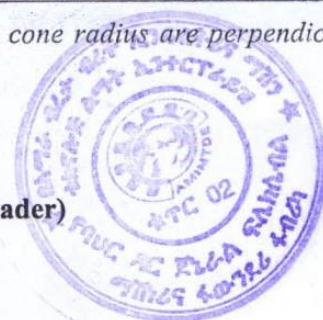
4.) Shaft angle is the sum of the pinion pitch cone angle and the gear pitch cone angle $\Sigma = \delta_1 + \delta_2$

5). $\delta a = \delta + \phi a$, where, δa = tip cone angle (face angle). at which the compound rest is swiveled.

ϕa = addendum angle

6). $\phi_f = \tan^{-1} \left(\frac{2.4 \cdot \sin \delta}{z} \right)$, Note: The dedendum angle of the pinion and the addendum angle of the gear should be same for mating. There for, $\phi_{fp} = \phi_{ag}$. The pitch cone radius are perpendicular with the dedendum line.

$$\tan \phi_f = \frac{\text{opp}}{\text{adj}} = \frac{hf}{R}, \text{ but. } hf = 1.2m$$

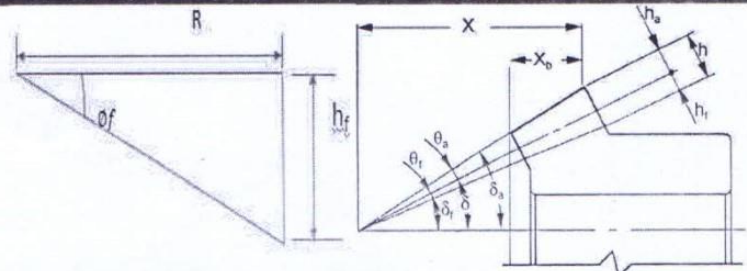


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and $R = \frac{m \cdot z}{2 \sin \theta}$, therefore.

$$\tan \phi_f = \frac{1.2 \cdot m}{\frac{m \cdot z}{2 \sin \theta}} = \frac{1.2 \cdot m \cdot 2 \sin \theta}{m \cdot z} = \frac{2.4 \cdot \sin \theta}{z}$$

Thus, $\phi_f = \tan^{-1} \left(\frac{2.4 \cdot \sin \theta}{z} \right)$

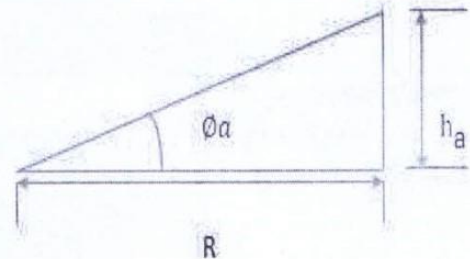


7). $\phi_a = \tan^{-1} \left(\frac{2 \cdot \sin \theta}{z} \right)$, **Note:** The addendum angle of the pinion and the dedendum angle of the gear should be same for mating. $\phi_{a_p} = \phi_{f_g}$. The pitch cone radius are perpendicular with the addendum line.

$\tan \phi_a = \frac{\text{opp}}{\text{adj}} = \frac{h_a}{R}$, but $h_a = m$, and $R = \frac{m \cdot z}{2 \sin \theta}$, therefore

$$\tan \phi_a = \frac{h_a}{R} = \frac{m}{\frac{m \cdot z}{2 \sin \theta}} = \frac{1.2 \cdot m \cdot 2 \sin \theta}{m \cdot z}$$

Thus, $\phi_a = \tan^{-1} \left(\frac{2 \cdot \sin \theta}{z} \right)$



8). $x = R \cdot \cos \theta - h_a \cdot \sin \theta$. Where, x = Pitch apex to crown, R = radius length of cone up to vertex (Pitch cone distance), h_a = addendum and θ = pitch cone angle.

9). $x_b = \frac{b \cdot \cos \theta_a}{\cos \phi_a}$. Where, x_b = axial face width, b = bevel gear tooth length, θ_a = tip cone angle (face angle) and ϕ_a = addendum angle.

10). $b \leq \frac{1}{3} \cdot R$, where, b = bevel gear tooth length, R = radius length of cone up to vertex (Pitch cone distance).

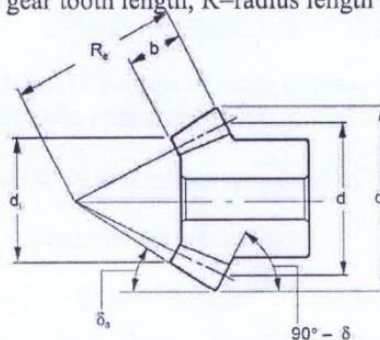


Fig: Dimensions and Angles of Bevel Gear

11). $d_i = D - \frac{2 \cdot b \cdot \sin \theta_a}{\cos \phi_a}$. Where, d_i = Inner outside diameter D = outside diameter of the bevel gear, b = bevel gear tooth length, θ_a = tip cone angle (face angle) and ϕ_a = addendum angle.

12). $d_f = d - (1.157 \cdot 2 \cdot m \cdot \cos \theta)$. where, d_f = root diameter, d = pitch diameter, m = module, θ = pitch cone angle.

13). $Q = 90^\circ - \theta$ where, Q = back cone angle.

14). $h = 2.2m$ where, h = working depth, m = modules.

15). $P = \pi \cdot m$, where, m = module.

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16). $h_a = m$, where, h_a = addendum.

17). $h_f = 1.2 \cdot m$, where, h_f = dedendum, m = module.

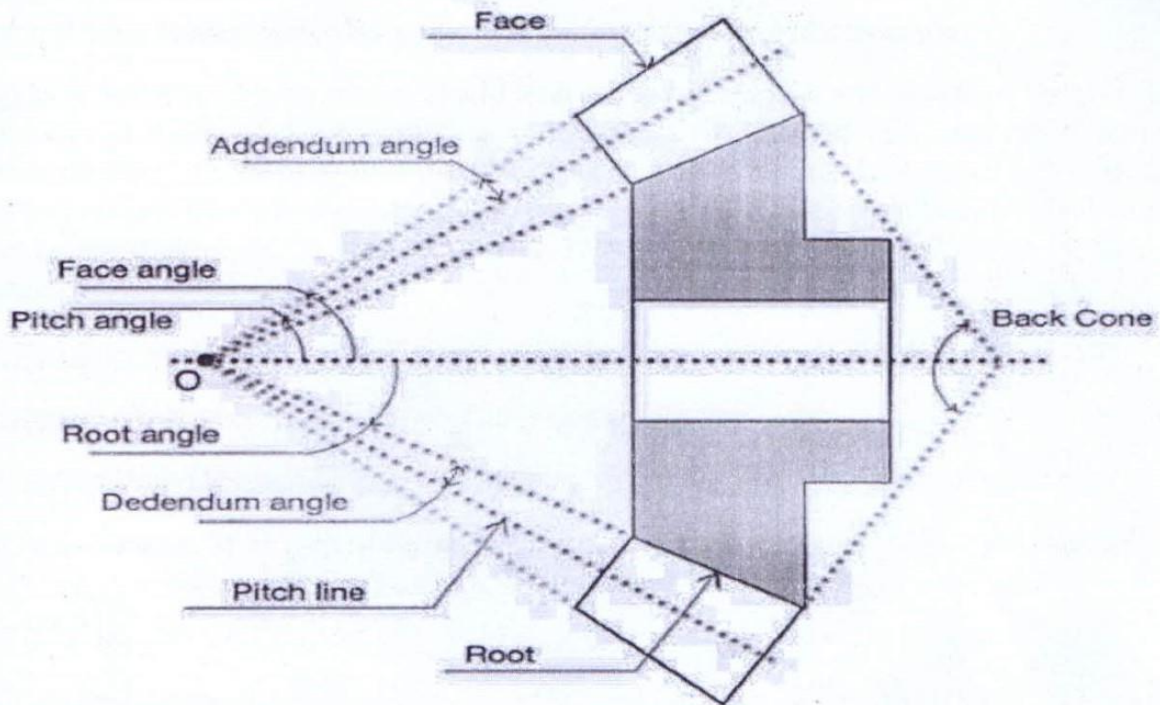


Fig: nomenclature of bevel gear.

Ex1:- If the straight bevel gear having 15 number of teeth, cutting module 3mm and its pitch cone angle is 15° . Now calculate the necessary data so as to manufacture the given bevel gear.

Given Data: $m = 3\text{mm}$, $\delta = 15^\circ$, $z = 15$: Now calculate the necessary data to manufacture the given bevel gears.

Solutions: Calculate the outside diameter of the work piece:

$$D_a = m(z + 2 \cos \delta) = 3(15 + 2 \cos 15^\circ) = \underline{50.8\text{mm}} \text{ (this is the max diameter).}$$

Calculate the root angle of the bevel gear, $\delta_r = \delta_f + \phi_f$, $\delta_f = \delta - \phi_f = 15^\circ - \tan^{-1}(2.4 \cdot \sin 15^\circ)/15 = 15 - 2.37 = \underline{12.63^\circ}$ (in which the dividing head angle is swiveled).

Calculate the back cone angle of the given bevel gear, $Q = 90^\circ - \delta$, $90^\circ - 15^\circ = \underline{75^\circ}$

Calculate the tip cone angle of the given bevel gears, $\delta_a = \delta + \phi_a = 15^\circ + \tan^{-1}(2 \cdot \sin 15^\circ)/15 = 15^\circ + 1.97^\circ = \underline{16.97^\circ}$ (in which the compound rest angle is swiveled).

Calculate the radius length of cone for a given bevel gear, $R = \frac{mz}{2 \sin \delta}$, $R = 3 \cdot 15 / 2 \sin 15^\circ = \underline{17.59\text{mm}}$

Calculate the bevel gear tooth length, $b \leq 1/3 \cdot R$, $b \leq 1/3 \cdot 17.59$, $b \leq \underline{5.86\text{mm}}$.

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Crank movement (T) = $\frac{40}{N} = \frac{40}{15} = 2\frac{10}{15}$, since 15 plate number are not supplied with standard set of plate number. Multiply both the nominator and denominator by 2 to get the standard set of plate. i.e. $2\frac{10 \times 2}{15 \times 2} = 2\frac{20}{30}$, thus the index crank should be 2 complete turn and moved 20 holes on a 30 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Ex2:- A pair of bevel gears consists of a 30 teeth pinion meshing with a 48 teeth gear. The gears are mounted on shafts; which are intersecting at right angles. The module at the large end of tooth is 4mm calculate? A). the pitch circle diameter of the pinion and the gear. B).the pitch cone angle for pinion and gear. C). the cone distance. D). Crank movement (T). The available index plate hole circle are as follow: plate1: 24, 25, 28, 30, 34, Plate2: 37, 38, 39, 41, 42, 43 Plate3: 46, 47, 49, 51, 53, and plate4: 54, 57, 58, 59, 62, 66.

Given data: $z_p=30$, $z_g=48$, $m=4$. The gear are intersecting at right angle (the shaft angle are 90°).

Solution: A). the pitch circle diameter of the pinion and the gear.

$d_p = m \cdot z_p = 4 \cdot 30 = \underline{120\text{mm}}$, $d_g = m \cdot z_g = 4 \cdot 48 = \underline{192\text{mm}}$, B).the pitch cone angle for pinion and gear.

If the shaft angle is 90° the pitch cone angle should be calculated as follow $\theta_p = \tan^{-1}(\frac{z_p}{z_g})$, and $\theta_g = \tan^{-1}(\frac{z_g}{z_p})$.

$$\theta_p = \tan^{-1}(\frac{30}{48}) = \underline{32^\circ}, \theta_g = \tan^{-1}(\frac{48}{30}) = \underline{58^\circ}.$$

C). the cone distance $R_p = \frac{m \cdot z_p}{2 \sin \theta_p} = \frac{4 \cdot 30}{2 \sin 32} = \frac{120}{1.0598} = \underline{113.2}$ and $R_g = \frac{m \cdot z_g}{2 \sin \theta_g} = \frac{4 \cdot 48}{2 \sin 58} = \frac{192}{1.696} = \underline{113.2}$

NB: the cone distance of the pinion and the gear should be same.

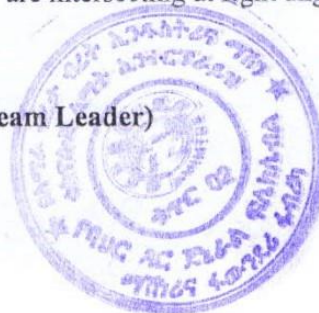
D). Crank movement of the pinion (T_p). $T_p = \frac{40}{N} = \frac{40}{30} = 1\frac{10}{30}$, thus the index crank should be 1 complete turn and moved 10 holes on a 30 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Crank movement of the gear (T_g). $T_g = \frac{40}{N} = \frac{40}{48}$, Since 48 plate number are not supplied with standard plate number. This means that less than one full turn is required for each division. $40/48$ must be reduced to its lowest term, divide both the nominator and denominator by 2. The lowest term which is $\frac{20}{24}$, thus the index crank should be moved 20 holes on a 24 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Ex3:- A pair of straight bevel gear consist of 30 teeth meshing with a 45 teeth gear .Module and face widths are 6mm and 50mm respectively .calculate the following:

A) Pitch cone angle for the pinion and the gear. B) Cone distance. C) Face cone angle, pinion. D) Root cone angle, pinion. E) Crank movement, pinion.

Given data: $z_p=30$, $z_g=45$, $m=6$, $b=50\text{mm}$. The gear are intersecting at right angle (the shaft angle are 90°).



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Solution: A). the pitch circle diameter of the pinion and the gear. $d_p = m \cdot z_p = 6 \cdot 30 = 180 \text{ mm}$, $d_g = m \cdot z_g = 6 \cdot 45 = 270 \text{ mm}$, B). the cone distance $R_p = \frac{m \cdot z_p}{2 \sin \theta_p}$. But first calculate the pitch cone angle for pinion and gear. If the shaft angle is 90° the pitch cone angle should be calculated as follow $\theta_p = \tan^{-1}(\frac{z_p}{z_g})$, and $\theta_g = \tan^{-1}(\frac{z_g}{z_p})$. $\theta_p = \tan^{-1}(\frac{30}{45}) = 33.7^\circ$, $\theta_g = \tan^{-1}(\frac{45}{30}) = 56.3^\circ$. Therefore, $R_p = \frac{180}{2 \sin 33.7} = \frac{180}{1.10968} = 162.2 \text{ mm}$. Or by using pitch cone angle of the gear we can get the same answer. Therefore, $R_g = \frac{m \cdot z_g}{2 \sin \theta_g} = \frac{270}{2 \sin 56.3} = \frac{270}{1.6639} = 162.2 \text{ mm}$.

C). Face cone angle, pinion. $\theta_a = \theta + \phi_a$, where, $\theta_a = \text{tip cone angle (face angle)}$ at which the compound rest is swiveled. $\phi_a = \text{addendum angle}$. $\phi_a = \tan^{-1}(\frac{2 \cdot \sin \theta}{z}) = \tan^{-1}(\frac{2 \cdot \sin 33.7}{30}) = 2.12^\circ$. Therefore, $\theta_a = \theta + \phi_a = 33.7^\circ + 2.12^\circ = 35.82^\circ$.

D). Root cone angle, pinion. $\theta_f = \theta - \phi_f$, but $\phi_f = \tan^{-1}(\frac{2.4 \cdot \sin \theta}{z}) = \tan^{-1}(\frac{2.4 \cdot \sin 33.7}{30}) = \tan^{-1}(0.04438) = 2.54^\circ$. Therefore, $\theta_f = \theta - \phi_f = 33.7^\circ - 2.54^\circ = 31.16^\circ$.

E). Crank movement, pinion (T). $T = \frac{40}{N} = \frac{40}{30} = 1 \frac{10}{30}$, thus the index crank should be 1 complete turn and moved 10 holes on a 30 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.



Spiral Bevel Gear



A spiral bevel gear: is a bevel gear with helical teeth. Spiral bevel gears are a specialized type of bevel gears with unique design characteristics that allow them to transmit power between intersecting shafts at different angles. These gears are widely used in various applications due to their efficiency, precision, and ability to handle high loads.

Spiral bevel gears are primarily used to transmit rotational motion and power between non-parallel intersecting shafts. They are commonly found in applications where changes in speed, direction, or torque are required. Spiral bevel gears are especially suitable for high-speed, high-load, and precision applications, making them prevalent in automotive, aerospace, industrial machinery, and power transmission systems.

NB: Manufacturing process is same as helical gear.

In order to cut a spiral bevel gear, any two of the following must be known:-

- d) **The lead of the helix (L):-** it is the longitudinal distance the helix advances axially in one complete revolution of the work piece.
- e) **The angle of the helix (β):-** is performed by the intersection of the helix with the axis of the work piece.
- f) **The Pitch circle diameter (d) or circumference of the pitch circle diameter of work piece.**

Cutting a helix: to cut a helix on a cone, the following steps are necessary:-

- 3) Swing the table in proper direction to the angle of the helix.
- 4) The work must rotate one turn while the table travels lengthwise the distance equal to the lead. This is achieved by installing the proper change gears between the worm shaft on the dividing head, and on the milling machine lead screw.

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Determine the direction to swing the milling machine table:-

- If the helix slopes down to the right, **it is a right hand helix.** To cut right helix, moving the right end of the table towards the column or by moving it in a counterclockwise direction. If you use a compound gear train, **use zero or even number idler gears for right hand helix.** But if you use a simple gear train **use one or add number idler gears for right hand helix.**
- If the helix slopes down to the left, **it is left hand helix.** To cut left helix, moving the left end of the table towards the column or by moving it in a clockwise direction. If you use a compound gear train, **use one or odd number idler gears for left hand helix.** But if you use a simple gear train **use zero or even number idler gears for left hand helix.**

OR

- **A right hand spiral bevel gear:** is one in which the outer half of a tooth is inclined in the clockwise direction from the axial plane through the midpoint of the tooth as viewed by an observer looking at the face of the gear.
- **A left hand spiral bevel gear:** is one in which the outer half of a tooth is inclined in the counterclockwise direction from the axial plane through the midpoint of the tooth as viewed by an observer looking at the face of the gear.

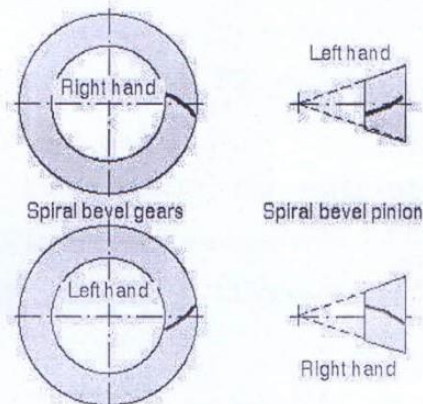


Fig: Direction of spiral bevel gear

The general formula for Spiral Bevel Gear will be explained as follow:

NB: The general formula of spiral bevel gear is same as straight bevel gear. The only deference is there is a helix angle on the spiral bevel gear and pitch circle diameter. The spiral angle is between **35° to 45°.**

1). $L = \frac{\pi d}{\tan \beta}$ where, L= lead of helix, d= pitch circle diameter of the work piece, β = helix angle (the milling machine swiveled).

2). $d = m_t * z$. where, d= pitch circle diameter, m_t =tangential module, z= number of teeth.

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But $m_t = \frac{mn}{\cos \beta}$ where, m_n = Normal module β = helix angle, Therefore, $d = \frac{mn}{\cos \beta} * z$

3). $D = d + 2m_t \cos \theta$, where, D = outside diameter of the spiral bevel gear, z = the number of gear tooth, m = module cutter, d = pitch diameter, θ = pitch cone angle

4). $P_t = \pi * m_t$ where, P_t = transvers (tangential) pitch, m_t = tangential module.

5). $P_n = \pi * m_n$ where, P_n = normal pitch, m_n = normal module. But $m_n = m_t * \cos \beta$.

Therefore, $P_n = \pi * m_t * \cos \beta = P_t * \cos \beta$.

Ex1:- A spiral bevel gear having module of 4, number of teeth 16, the pitch cone angle 37° , face width (b) 19mm and helix angle 30° . Using a standard set of gears (25,30,35,40,50,55,60,70,80,90,100). The available index plate hole circle are as follow: plate1: 24, 25, 28, 30, 34, Plate2: 37, 38, 39, 41, 42, 43 Plate3: 46, 47, 49, 51, 53, and plate4: 54, 57, 58, 59, 62, 66. The milling machine lead screw has a pitch of 5mm. Calculate?

- A). pitch circle diameter. B). Outside diameter. C). Cone distance. D). Root diameter (d_f).
E). Inner outside diameter (d_i). F). Total depth. G). Dedendum angle. H). Addendum angle.
I). Lead. J). Change gears (i). K). Crank movement (T)

Given data: $z=16$, $m=4$, $b=19$ mm, $\theta=37^\circ$ and $\beta=30^\circ$. The gear are intersecting at right angle (the shaft angle are 90°).

Solution: A). pitch circle diameter (d) = $m_t * z$, $m_t = \frac{mn}{\cos \beta}$, Therefore, $d = \frac{mn}{\cos \beta} * z = \frac{4}{\cos 30} * 16 = 73.9$ mm.

B). Outside diameter (D) = $d + 2m_t \cos \theta = 73.9 + 2 * 4 * \cos 37^\circ = 80.2$ mm. C). Cone distance (R) = $\frac{d}{2 \sin \theta} = \frac{73.9}{2 \sin 37} = 61.3$ mm. D). Root diameter (d_f) = $d_f = d - (1.157 * 2 * m * \cos \theta) = d_f = 73.9 - (1.157 * 2 * 4 * \cos 37) = 66.5$ mm. E). Inner outside diameter $d_i = D - \frac{2 * b * \sin \theta a}{\cos \theta a}$, but $\theta a = \theta + \phi a$. $\phi a = \tan^{-1}(\frac{2 * \sin \theta}{z}) = \tan^{-1}(\frac{2 * \sin 37}{16}) = 4.3^\circ$ $\phi f = \tan^{-1}(\frac{2.4 * \sin \theta}{z}) = \tan^{-1}(\frac{2.4 * \sin 37}{16}) = 5.16^\circ$. Therefore, $\theta a = \theta + \phi a = 37^\circ + 4.3^\circ = 41.3^\circ$.

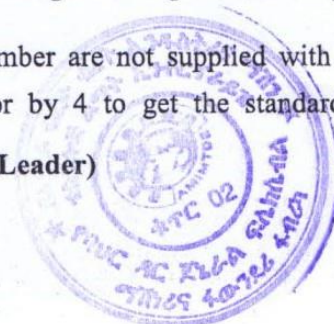
$d_i = D - \frac{2 * b * \sin \theta a}{\cos \theta a} = 80.2 - \frac{2 * 19 * \sin 41.3}{\cos 4.3} = 80.2 - \frac{25.08}{0.997} = 80.2 - 24.082 = 55.04$ mm.

F). Total depth (h) = $2.2m = 2.2 * 4 = 8.8$ mm

I). Lead (L) = $\frac{\pi d}{\tan \beta} = \frac{\pi * 73.9}{\tan 30} = 400$ mm.

J). Change gears (i) = $\frac{\text{product of driven gears}}{\text{product of driver gears}} = \frac{\text{lead of helix to be cut (mm)}}{\text{lead of machine (mm)}} = \frac{L}{p * 40} = \frac{400}{5 * 40} = \frac{400}{200} = \frac{2}{1}$ Since 2 and 1 tooth gear are not supplied with standard set of gear. It is necessary to multiply both the nominator and the denominator by a number that will suit the standard gear available. Therefore to get the standard set of gear multiply the nominator and the denominator by 50. Change gears = $\frac{\text{driven gears}}{\text{driver gears}} = \frac{2 * 50}{1 * 50} = \frac{100}{50}$, driven gear (attached on the dividing head worm shaft = 100) and driver gear (attached on milling machine lead screw = 50). In this case the gear arrangement is simple gear train.

K). Crank movement (T) = $\frac{40}{N} = \frac{40}{16} = 2\frac{4}{8}$. Since 8 plate number are not supplied with standard plate number. Multiply both the nominator and denominator by 4 to get the standard set of plate.



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$T = 2 \frac{4+4}{8+4} = 2 \frac{16}{24}$. Thus the index crank should be 2 complete turn and moved 16 holes on a 24 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

Sprocket Gear

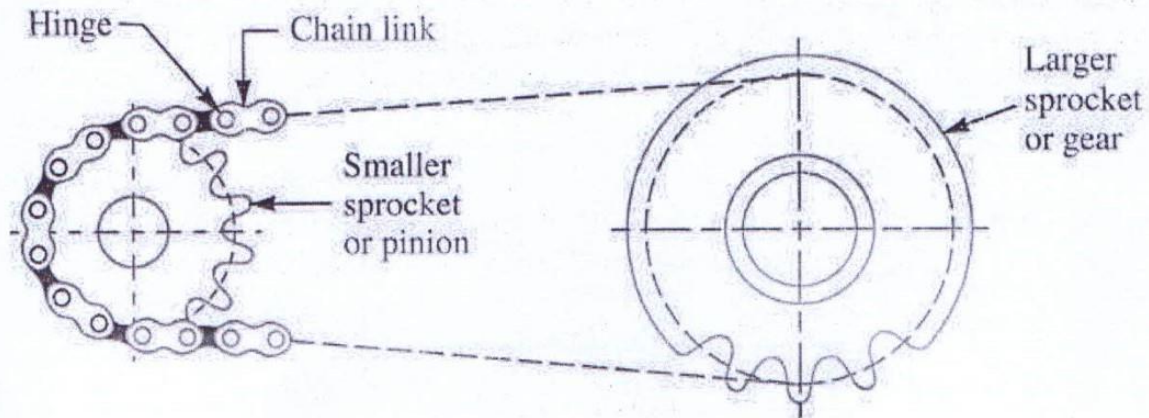


Fig: Sprockets and chain.

The chains are mostly used to transmit motion and power from one shaft to another, when the center distance between their shafts is short such as in bicycles, motor cycles, agricultural machinery, conveyors, rolling mills, road rollers etc. The chains may also be used for long center distance of up to 8 meters. The chains are used for velocities up to 25 m/s and for power up to 110 kW. In some cases, higher power transmission is also possible.

Terms Used in Chain Drive

a). **Pitch of chain.** It is the distance between the hinge center of a link and the corresponding hinge center of the adjacent link, as shown in bellow. It is usually denoted by p .

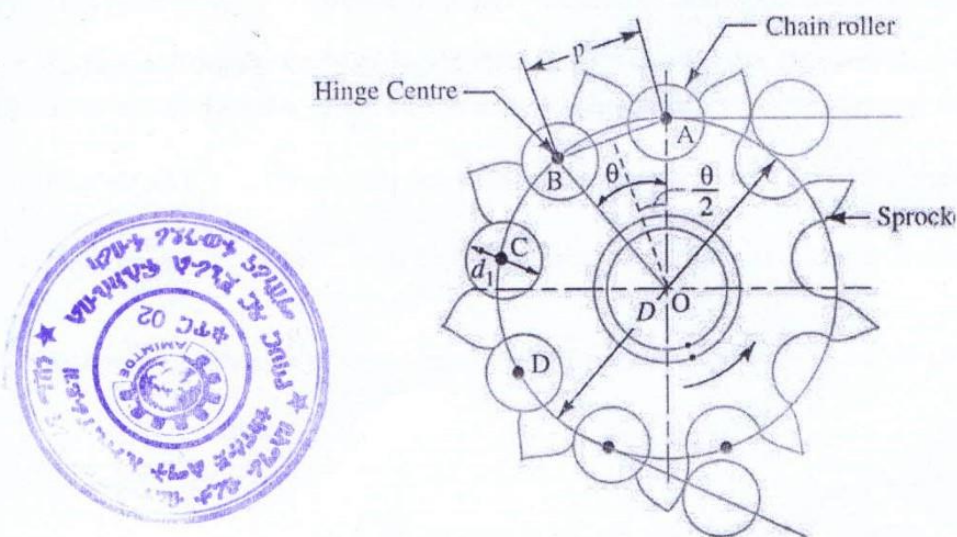


Fig: Terms used in chain drive.

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b). **Pitch circle diameter of chain sprocket.** It is the diameter of the circle on which the hinge centers of the chain lie, when the chain is wrapped round a sprocket as shown in the above figure. The points A, B, C, and D are the hinge centers of the chain and the circle drawn through these centers is called pitch circle and its diameter (d) is known as pitch circle diameter.

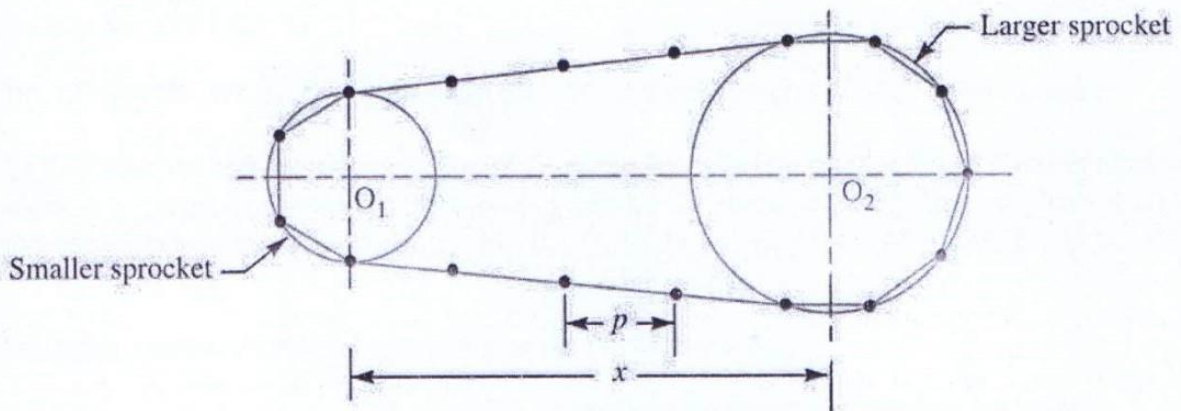


Fig: Length of chain.

c). The length of the chain (L) must be equal to the product of the number of chain links (K) and the pitch of the chain (p).

The general formula for Sprocket Gear will be explained as follow:

$$1). x = \frac{p}{4} \left[K - \frac{T_1 + T_2}{2} + \sqrt{\left(K - \frac{T_1 + T_2}{2} \right)^2 - 8 \left(\frac{T_2 - T_1}{2\pi} \right)^2} \right]$$

Where, x = center distance between the two gears, p = pitch of the chain, T_1 = number of tooth on small sprocket gear, T_2 = number of tooth on large sprocket gear and K = number of chain link.

2). $L = k \cdot p$ where, L = length of chain, k = number of chain link and p = pitch of the chain.

3). $D_o = d + 0.9d_r$ for $T > 19$ or $D_o = d + 0.8d_r$ for $T < 18$ Where, D_o = outside diameter of sprocket gear, d = reference diameter of sprocket gear / pitch diameter/, d_r = roller diameter and T = number of tooth.

4). $p = d \cdot \sin\left(\frac{\theta}{2}\right)$ Where, p = pitch of the chain, d = pitch circle diameter, θ = angle of tooth sprocket.

5). $d = \frac{p}{\sin\left(\frac{\theta}{2}\right)}$ Where, p = pitch of the chain, d = pitch circle diameter, θ = angle of tooth sprocket.

6). $\theta = \frac{360^\circ}{T}$ θ = angle of tooth sprocket and T = number of tooth. Therefore $d = \frac{p}{\sin\left(\frac{\theta}{2}\right)} = d = \frac{p}{\sin\left(\frac{360^\circ}{2T}\right)}$

$$d = \frac{p}{\sin\left(\frac{180^\circ}{T}\right)}$$

7). $h = 0.95 \cdot d_r$ Where, h = depth of cut and d_r = roller diameter.

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8). $d_f = d - d_r$ where, d_f = root diameter of sprocket gear, d = pitch circle diameter and d_r = roller diameter.

9). $K = \frac{T_1 + T_2}{2} + \frac{2x}{p} + \left[\frac{T_2 - T_1}{2\pi} \right]^2 \frac{p}{x}$ Where, K = number of chain link, x = center distance between the two gears, p = pitch of the chain, T_1 = number of tooth on small sprocket gear, T_2 = number of tooth on large sprocket gear.

10). $G = \left(\frac{0.1}{0.2} \right) * d_r$ Where, G = gap between the roller and sprocket groove and d_r = roller diameter.

Ex1:- If the pitch of the sprocket gear has 18.5mm, the number of the tooth is 35 and diameter of roller diameter is 10mm, now calculate the necessary data for the sprocket gears? Use a standard set plate number given below: A. (24, 25, 28, 30, 34,) B. (37, 38, 39, 41, 42, 43) C. (46, 47, 49, 51, 53,) D. (54, 57, 58, 59, 62, 66)

Solution: Outside diameter of Sprocket gear can be calculated as:

$D_o = d + 0.9d_r$, where D_o , is tip diameter of the sprocket gears.

$d = \frac{P}{\sin \theta}$, where d is the reference diameter of sprocket gear.

$$\text{But } \theta = \frac{360^\circ}{T} = \frac{360}{35} = 10.28^\circ$$

$$d = \frac{18.5}{\sin 10.28} = 103.6\text{mm}$$

$$D_o = d + 0.9d_r = 103.6 + 0.9 * 10 = 112.6\text{mm}$$

Root diameter of the sprocket gear can be calculated as:

$$d_f = d - d_r = 103.6 - 10 = 93.6\text{mm}$$

Depth of cut can be calculated as:

$$h = 0.95 * d_r = 0.95 * 10 = 9.5\text{mm}$$

Crank movement can be calculated as:

$\text{Crank} = \frac{40}{T} = \frac{40}{35} = \frac{8}{7} = 1\frac{1}{7}$, since 7 plate number are not supplied with standard plate number listed the above. Multiply both the nominator and denominator by 4 to get the standard set of plate. i.e. $1\frac{1*4}{7*4} = 1\frac{4}{28}$, thus the index crank should be 1 complete turn and moved 4 holes on a 28 hole circle for each division. When counting holes, start with the first hole ahead of the index pin.

NB: The tool used to manufacture sprocket gear are pitch cutter



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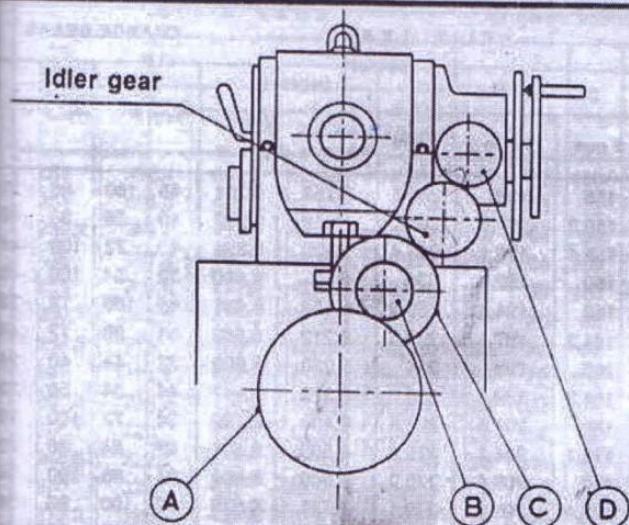
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UNIVERSAL DIVIDING HEAD

TABLE FOR SPIRAL MILLING



Worm gear ratio. 1 : 40

Change gears: 24, 24, 28, 32, 40, 44,
48, 56, 64, 72, 86, 100.

HELIX HAND

Right hand: With no or two idler gears

Left hand: With one idler gear

HELIX LEAD					CHANGE GEARS				HELIX LEAD					CHANGE GEARS			
MM			INCHES		A	B	C	D	MM			INCHES		A	B	C	D
5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm					5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm				
13,4	16,1	21,4	0,670	0,536	86	24	100	24	55	66	88	2,750	2,200	64	40	100	44
15,6	18,7	25	0,781	0,624	86	24	100	28	56	67,2	89,6	2,800	2,240	24	24	100	28
16	19,2	25,6	0,800	0,640	72	24	100	24	57	68,4	91,2	2,849	2,279	64	28	86	56
17,9	21,4	28,6	0,893	0,714	86	24	100	32	58,2	69,8	93,1	2,909	2,327	44	32	100	40
18,6	22,3	29,8	0,930	0,744	72	24	86	24	59,2	71	94,7	2,960	2,368	44	28	86	40
21	25,2	33,6	1,050	0,840	64	24	100	28	60	72	96	3	2,400	40	24	56	28
22,3	26,8	35,7	1,116	0,892	86	24	100	40	61,1	73,3	97,8	3,055	2,444	44	28	100	48
24	28,8	38,4	1,200	0,960	48	24	100	24	62	74,4	99,2	3,101	2,480	72	40	86	48
25	30	40	1,250	1	64	24	72	24	63	75,6	100,8	3,150	2,520	100	28	64	72
26	31,2	41,7	1,302	1,041	86	28	100	40	64	76,8	102,4	3,200	2,560	100	28	56	64
27,4	32,9	43,9	1,371	1,096	56	24	100	32	65,1	78,1	104,2	3,256	2,604	24	24	86	28
28	33,6	44,8	1,400	1,120	48	24	100	28	66	79,2	105,6	3,300	2,640	64	44	100	48
29,2	35	46,7	1,458	1,166	64	24	72	28	67	80,4	107,2	3,349	2,679	86	40	100	72
30	36	48	1,500	1,200	64	24	100	40	68,1	81,7	108,9	3,403	2,722	64	28	72	56
31	37,2	49,6	1,550	1,240	72	24	86	40	68,8	82,5	110	3,438	2,750	48	24	84	44
32	38,4	51,2	1,600	1,280	48	24	100	32	70	84	112	3,500	2,800	64	40	100	56
33	39,6	52,8	1,650	1,320	64	24	100	44	71	85,2	113,7	3,552	2,841	44	56	86	24
34,1	40,9	54,6	1,706	1,364	72	24	86	44	72	86,4	115,2	3,600	2,880	48	72	100	24
35	42	56	1,750	1,400	64	28	100	40	73,1	87,8	117	3,657	2,925	56	64	100	32
36	43,2	57,6	1,800	1,440	64	24	100	48	74,1	88,9	118,5	3,704	2,963	48	32	72	40
37,2	44,6	59,5	1,860	1,488	56	28	86	32	75	90	120	3,750	3	32	24	48	24
38,1	45,7	61	1,905	1,524	56	24	72	32	76	91,2	121,6	3,799	3,039	48	56	86	28
39,1	46,9	62,5	1,954	1,563	40	24	86	28	77	92,4	123,2	3,850	3,080	64	44	100	56
40	48	64	2	1,600	40	24	72	24	78,2	93,9	125,2	3,911	3,128	72	44	100	64
41,1	49,4	65,8	2,057	1,645	28	24	100	24	79,5	95,4	127,3	3,977	3,181	44	28	64	40
42	50,4	67,2	2,100	1,680	64	24	100	56	80	96	128	4	3,200	40	24	48	32
43,4	52,1	69,5	2,171	1,736	72	24	86	56	81,2	97,4	129,9	4,060	3,248	44	64	86	24
44	52,8	70,4	2,200	1,760	48	24	100	44	82,3	98,7	131,6	4,114	3,291	28	48	100	24
45	54	72	2,250	1,800	40	24	64	24	82,9	99,5	132,6	4,144	3,315	44	56	86	28
46,5	55,8	74,4	2,326	1,860	64	32	86	40	84	100,8	134,4	4,200	3,360	64	48	100	56
47,4	56,8	75,8	2,368	1,894	44	28	86	32	85,1	102,1	136,1	4,253	3,402	56	64	86	32
48	57,6	76,8	2,400	1,920	56	28	100	48	86	103,2	137,6	4,300	3,440	56	86	100	28
49	58,8	78,4	2,450	1,960	64	28	100	56	87,2	104,7	139,6	4,361	3,488	64	100	86	24
50	60	80	2,500	2	48	24	56	28	88	105,6	140,8	4,400	3,520	24	24	100	44
51,2	61,4	81,9	2,558	2,046	64	32	86	44	89,3	107,2	142,9	4,466	3,572	40	48	86	32
52,1	62,5	83,4	2,605	2,084	40	28	86	32	90	108	144	4,500	3,600	64	72	100	40
53,2	63,8	85,1	2,658	2,126	56	32	86	40	91	109,2	145,5	4,548	3,638	72	44	86	64
54	64,8	86,4	2,700	2,160	64	24	100	72	93	111,6	148,8	4,651	3,720	24	40	86	24

UNIVERSAL DIVIDING HEAD

TABLE FOR SPIRAL MILLING

HELIX LEAD					CHANGE GEARS				HELIX LEAD					CHANGE GEARS			
MM			INCHES		A	B	C	D	MM			INCHES		A	B	C	D
5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm					5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm				
94,3	113,1	150,8	4,714	3,771	40	44	56	24	155	186	248,1	7,752	6,201	48	100	86	32
95,2	114,3	152,4	4,762	3,809	28	40	72	24	156,3	187,6	250,1	7,815	6,252	40	56	86	48
96	115,2	153,6	4,800	3,840	24	48	100	24	158,4	190,1	253,4	7,920	6,336	40	72	100	40
97	116,4	155,2	4,849	3,879	44	32	72	48	160	192	256	8	6,400	32	64	100	44
98	117,6	156,8	4,900	3,920	32	56	100	28	162	194,4	259,3	8,102	6,481	48	100	72	28
99	118,8	158,4	4,950	3,960	44	56	72	28	164,2	197,1	262,6	8,212	6,569	64	86	72	44
100	120	160	5	4	24	24	56	28	165	198	264	8,250	6,600	32	44	40	24
102,1	122,5	163,4	5,105	4,084	48	28	64	56	166,2	199,5	266	8,312	6,649	44	64	56	32
103,1	123,8	165	5,156	4,124	32	44	64	24	168	201,6	268,8	8,400	6,720	24	72	100	28
104,2	125	166,7	5,210	4,168	40	64	86	28	170,1	204,1	272,2	8,506	6,804	28	64	86	32
105	126	168	5,250	4,200	32	24	40	28	172	206,4	275,2	8,600	6,880	24	86	100	24
106,1	127,3	169,7	5,303	4,242	44	28	48	40	174,4	209,3	279,1	8,721	6,976	32	100	86	24
107	128,4	171,1	5,348	4,278	32	44	72	28	175	210	280	8,750	7	24	28	32	24
108	129,6	172,8	5,400	4,320	32	72	100	24	176	211,2	281,6	8,800	7,040	24	48	100	44
109,1	130,9	174,6	5,455	4,364	44	48	56	28	178,2	213,8	285,1	8,909	7,127	40	56	44	28
110	132	176	5,500	4,400	40	44	48	24	180	216	288	9	7,200	32	48	40	24
111,1	133,3	177,8	5,556	4,444	24	40	72	24	182,3	218,8	291,7	9,115	7,292	48	100	64	28
112	134,4	179,2	5,600	4,480	24	56	100	24	184,2	221	294,7	9,210	7,368	40	72	86	44
113,1	135,8	181	5,657	4,525	44	56	72	32	185,2	222,2	296,3	9,260	7,408	48	100	72	32
114	136,8	182,3	5,698	4,558	32	56	86	28	186	223,2	297,7	9,302	7,441	24	48	86	40
115,2	138,2	184,3	5,760	4,608	40	72	100	32	188,1	225,7	301	9,406	7,524	40	86	64	28
116,4	139,6	186,2	5,818	4,654	44	64	100	40	190,5	228,6	304,8	9,524	7,619	28	40	48	32
117	140,4	187,1	5,848	4,678	28	44	86	32	192	230,4	307,1	9,598	7,678	56	86	64	40
118,2	141,9	189,2	5,912	4,729	64	86	100	44	195,4	234,4	312,6	9,768	7,814	48	72	86	56
119	142,8	190,5	5,952	4,761	56	100	72	24	196,4	235,7	314,3	9,822	7,857	32	44	56	40
120	144	192	6	4,800	40	48	56	28	198	237,6	316,8	9,900	7,920	32	72	100	44
121,2	145,5	194	6,061	4,848	44	40	48	32	200	240	320	10	8	28	56	48	24
122,2	146,7	195,6	6,112	4,889	24	24	72	44	202	242,4	323,2	10,101	8,080	44	100	72	32
123,2	147,8	197,1	6,160	4,928	40	56	100	44	204,2	245	326,7	10,209	8,167	24	56	64	28
124	148,8	198,5	6,202	4,961	24	40	86	32	205,3	246,4	328,5	10,267	8,213	24	56	100	44
125	150	200	6,250	5	24	24	64	40	206,2	247,5	330	10,312	8,249	32	48	64	44
126	151,2	201,6	6,300	5,040	32	72	100	28	208,3	250	333,3	10,417	8,333	32	100	72	24
127,3	152,7	203,6	6,364	5,091	44	56	48	24	210	252	336	10,500	8,400	32	56	40	24
128	153,6	204,8	6,400	5,120	24	64	100	24	212,1	254,5	339,4	10,606	8,484	44	56	48	40
129	154,8	206,4	6,450	5,160	64	86	100	48	214,3	257,1	342,8	10,714	8,571	32	48	56	40
130,2	156,3	208,4	6,512	5,209	24	56	86	24	215	258	344	10,750	8,600	40	86	48	24
131	157,2	209,5	6,548	5,238	48	44	56	40	216	259,2	345,6	10,800	8,640	32	72	100	48
132	158,4	211,2	6,600	5,280	32	48	100	44	218,3	261,9	349,2	10,913	8,730	56	100	72	44
133,3	160	213,3	6,667	5,333	48	64	56	28	220	264	352	11	8,800	24	44	40	24
134	160,8	214,3	6,698	5,358	40	72	86	32	222,2	266,7	355,6	11,111	8,888	24	40	48	32
135	162	216	6,750	5,400	40	72	64	24	224	268,8	358,4	11,200	8,960	24	56	100	48
136,1	163,3	217,8	6,806	5,444	32	56	72	28	225	270	360	11,250	9	24	72	64	24
137,1	164,6	219,4	6,857	5,485	28	32	40	24	226,3	271,5	362	11,313	9,050	44	64	72	56
138,9	166,7	222,2	6,944	5,555	48	100	72	24	228	273,6	364,8	11,401	9,120	44	86	48	28
140	168	224	7	5,600	24	28	40	24	230,4	276,5	368,7	11,520	9,216	40	72	100	64
141,4	169,7	226,3	7,071	5,656	44	56	72	40	232,6	279,1	372,1	11,629	9,303	24	100	86	24
142,2	170,7	227,6	7,111	5,688	40	64	72	32	234,4	281,3	375,1	11,721	9,376	40	72	86	56
143,2	171,8	229,1	7,159	5,727	44	72	64	28	235,1	282,2	376,2	11,757	9,405	32	86	64	28
144	172,8	230,4	7,200	5,760	24	72	100	24	236,5	283,8	378,4	11,825	9,460	32	86	100	44
145,4	174,4	232,6	7,268	5,814	64	100	86	40	238,1	285,7	381	11,905	9,524	28	100	72	24
146,2	175,4	233,9	7,310	5,848	28	44	86	40	240	288	384	12	9,600	24	48	40	24
147,4	176,9	235,9	7,372	5,897	28	86	100	24	242,4	290,9	387,9	12,121	9,696	24	40	44	32
148	177,6	236,8	7,400	5,920	44	100	86	28	244,4	293,3	391,1	12,222	9,777	24	44	48	32
149,3	179,2	238,9	7,465	5,972	64	86	72	40	245	294	392	12,250	9,800	32	56	40	28
150	180	240	7,500	6	24	48	64	24	246,4	295,6	394,2	12,318	9,854	48	86	64	44
152	182,4	243,2	7,601	6,080	44	86	72	28	248,1	297,7	396,9	12,403	9,922	24	64	86	40
154	184,8	246,4	7,700	6,160	32	56	100	44	250	300	400	12,500	10	24	40	32	24

UNIVERSAL DIVIDING HEAD

TABLE FOR SPIRAL MILLING

HELIX LEAD					CHANGE GEARS				HELIX LEAD					CHANGE GEARS			
MM			INCHES		A	B	C	D	MM			INCHES		A	B	C	D
5 mm	6 mm	8 mm	4 TPI 6.35 mm	5 TPI 5.08 mm					5 mm	6 mm	8 mm	4 TPI 6.35 mm	5 TPI 5.08 mm				
252	302.4	403.2	12,600	10,080	32	72	100	56	365.5	438.6	584.7	18,273	14,618	28	100	86	44
254	304.8	406.3	12,698	10,158	28	64	72	40	366.7	440	586.7	18,333	14,666	28	56	48	44
255.2	306.2	408.3	12,758	10,206	28	64	86	48	368.6	442.3	589.7	18,428	14,742	28	86	40	24
256	307.2	409.6	12,800	10,240	28	64	100	56	370.4	444.5	592.6	18,519	14,815	24	100	72	32
258	309.6	412.8	12,900	10,320	32	86	100	48	372.1	446.5	595.4	18,605	14,884	40	100	86	64
260.5	312.6	416.8	13,024	10,419	24	56	86	48	374	448.8	598.4	18,700	14,960	44	72	56	64
262.5	315	420	13,125	10,500	32	72	48	28	375	450	600	18,750	15	32	100	40	24
264	316.8	422.4	13,200	10,560	24	72	100	44	376.2	451.5	602	18,812	15,049	32	86	40	28
265.2	318.2	424.3	13,258	10,606	44	100	48	28	378.8	454.5	606	18,939	15,151	44	100	48	40
266.7	320	426.7	13,333	10,666	24	64	48	24	380.6	456.7	608.9	19,029	15,223	44	100	86	72
268.8	322.5	430	13,438	10,750	24	86	64	24	382.2	458.7	611.6	19,111	15,288	40	86	72	64
270	324	432	13,500	10,800	32	72	40	24	384	460.8	614.4	19,200	15,360	24	72	100	64
272.2	326.7	435.6	13,611	10,888	24	56	48	28	385	462	616	19,250	15,400	32	56	40	44
274.3	329.1	438.8	13,713	10,970	40	64	56	48	388.9	466.7	622.2	19,444	15,555	24	40	24	28
275	330	440	13,750	11	24	44	32	24	390.6	468.7	625	19,531	15,624	32	100	64	40
277.8	333.3	444.4	13,889	11,111	24	100	72	24	392.7	471.2	628.3	19,635	15,708	40	72	44	48
280	336	448	14	11,200	24	56	40	24	394.2	473	630.7	19,710	15,768	40	86	48	44
282.9	339.4	452.5	14,142	11,313	40	72	56	44	396.8	476.2	634.9	19,840	15,872	28	100	72	40
284.1	340.9	454.5	14,204	11,363	44	100	64	40	398.2	477.8	637.1	19,908	15,926	24	86	72	40
285.2	342.2	456.3	14,260	11,408	24	56	72	44	400	480	640	20	16	24	72	48	32
286.4	343.7	458.3	14,322	11,457	48	100	64	44	404	484.8	646.4	20,200	16,160	44	100	72	64
288	345.6	460.8	14,400	11,520	24	72	100	48	408.2	489.8	653.1	20,410	16,328	28	100	56	32
294.9	353.8	471.8	14,743	11,794	28	86	100	48	412.6	495.1	660.2	20,630	16,504	32	72	48	44
296	355.2	473.6	14,800	11,840	44	100	86	56	414.8	497.8	663.7	20,740	16,592	24	64	72	56
300	360	480	15	12	24	48	32	24	415.6	498.7	665	20,780	16,624	28	64	44	40
303	363.6	484.8	15,150	12,120	44	100	48	32	416.6	499.9	666.6	20,830	16,664	32	100	72	48
304	364.8	486.5	15,202	12,161	44	86	72	56	418	501.6	668.8	20,900	16,720	32	86	72	56
305.4	366.5	488.7	15,272	12,217	40	56	44	48	420	504	672	21	16,800	32	56	40	48
306.1	367.3	489.8	15,306	12,244	28	100	56	24	422.4	506.9	675.8	21,120	16,896	32	86	56	44
308.6	370.3	493.7	15,429	12,343	40	72	56	48	426.4	511.7	682.2	21,320	17,056	24	100	86	44
310.1	372.1	496.1	15,504	12,403	48	100	86	64	428.6	514.3	685.8	21,430	17,144	40	100	56	48
312.5	375	500	15,625	12,500	24	100	64	24	430	516	688	21,500	17,200	24	86	40	24
314.3	377.1	502.8	15,714	12,571	24	44	28	24	436.4	523.7	698.2	21,820	17,456	44	72	48	64
315	378	504	15,750	12,600	32	72	40	28	438	525.6	700.8	21,900	17,520	24	86	72	44
320	384	512	16	12,800	24	64	40	24	440	528	704	22	17,600	32	64	40	44
322.5	387	516	16,125	12,900	32	86	40	24	442.2	530.6	707.5	22,110	17,688	28	86	100	72
324.1	388.9	518.5	16,204	12,963	24	100	72	28	444.4	533.3	711	22,220	17,776	40	100	72	64
325.6	390.7	521	16,280	13,024	40	100	86	56	446.8	536.2	714.9	22,340	17,872	44	86	56	64
326.5	391.8	522.5	16,327	13,061	28	64	56	40	448	537.6	716.8	22,400	17,920	32	86	48	40
328.5	394.2	525.5	16,423	13,138	32	86	72	44	450	540	720	22,500	18	24	72	64	48
330	396	528	16,500	13,200	40	72	48	44	454.6	545.5	727.4	22,730	18,184	24	100	44	24
332.2	398.7	531.6	16,612	13,289	28	100	86	40	456	547.2	729.6	22,800	18,240	48	86	44	56
334.4	401.3	535.1	16,722	13,377	40	86	72	56	458.4	550.1	733.4	22,920	18,336	40	100	48	44
335	402	536.1	16,752	13,401	44	86	56	48	460.8	553	737.3	23,040	18,432	56	86	48	72
336	403.2	537.6	16,800	13,440	24	72	100	56	462.8	555.4	740.5	23,140	18,512	24	100	72	40
338.3	405.9	541.2	16,914	13,531	44	100	86	64	465.2	558.2	744.3	23,260	18,608	32	100	86	64
340.9	409.1	545.4	17,045	13,636	32	100	44	24	466.6	559.9	746.6	23,330	18,664	32	64	48	56
342	410.4	547.3	17,102	13,681	32	86	44	28	470.4	564.5	752.6	23,520	18,816	32	86	64	56
344	412.8	550.4	17,200	13,760	32	86	100	64	476.2	571.4	761.9	23,810	19,048	48	100	56	64
345.5	414.6	552.8	17,275	13,820	56	86	64	72	480	576	768	24	19,200	40	64	48	72
348.8	418.6	558.1	17,442	13,953	32	100	86	48	482.6	579.1	772.2	24,130	19,304	28	86	56	44
350	420	560	17,500	14	24	56	32	24	484.8	581.8	775.7	24,240	19,392	24	64	44	40
355.6	426.7	568.9	17,778	14,222	24	64	48	32	486.2	583.4	777.9	24,310	19,448	32	100	72	56
358.4	430	573.4	17,918	14,334	24	86	48	24	488.8	586.6	782.1	24,440	19,552	24	44	24	32
360	432	576	18	14,400	24	72	40	24	490.8	589	785.3	24,540	19,632	32	72	44	48
363.6	436.3	581.8	18,181	14,544	28	56	44	40	492.8	591.4	788.5	24,640	19,712	24	86	64	44
364.6	437.5	583.3	18,229	14,583	32	100	48	28	495	594	792	24,750	19,800	32	72	40	44

UNIVERSAL DIVIDING HEAD

TABLE FOR SPIRAL MILLING

HELIX LEAD					CHANGE GEARS				HELIX LEAD					CHANGE GEARS			
MM			INCHES		A	B	C	D	MM			INCHES		A	B	C	D
5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm					5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm				
498,6	598,3	797,8	24,930	19,944	28	64	44	48	684	820,8	1094,4	34,200	27,360	44	96	32	56
500	600	800	25	20	24	72	48	40	685,8	823	1097,3	34,290	27,432	48	72	28	64
502,8	603,4	804,5	25,140	20,112	28	64	40	44	691	829,2	1105,6	34,550	27,640	32	86	56	72
510,2	612,2	816,3	25,510	20,408	28	100	56	40	694,4	833,3	1111	34,720	27,776	24	100	48	40
512	614,4	819,2	25,600	20,480	28	86	48	40	698	837,6	1116,8	34,900	27,920	56	100	44	88
514,2	617	822,7	25,710	20,568	24	72	56	48	700	840	1120	35	28	24	72	48	56
516	619,2	825,6	25,800	20,640	24	86	100	72	702	842,4	1123,2	35,100	28,080	28	86	56	64
520,8	625	833,3	26,040	20,832	32	100	48	40	711,2	853,4	1137,9	35,560	28,448	24	64	24	32
525	630	840	26,250	21	32	72	48	56	714,2	857	1142,7	35,710	28,568	32	100	56	64
526,6	631,9	842,8	26,330	21,064	28	86	56	48	720	864	1152	36	28,800	32	72	40	64
530,4	636,5	848,6	26,520	21,216	44	100	48	56	727,2	872,6	1163,5	36,360	29,088	44	100	40	64
535,8	643	857,3	26,790	21,432	48	100	56	72	733,4	880,1	1173,4	36,670	29,336	24	48	24	44
540	648	864	27	21,600	32	72	40	48	737,2	884,6	1179,5	36,860	29,488	28	86	40	48
542,8	651,1	868,2	27,130	21,704	24	100	86	56	740,8	889	1185,3	37,040	29,632	24	100	72	64
544,4	653,3	871	27,220	21,776	24	56	24	28	746,8	895,9	1194,6	37,330	29,864	32	100	72	86
545,4	654,5	872,6	27,270	21,816	40	100	44	48	748	897,6	1196,8	37,400	29,920	28	72	44	64
546	655,2	873,6	27,300	21,840	28	86	72	64	750	900	1200	37,500	30	48	100	40	72
548,6	658,3	877,8	27,430	21,944	28	64	40	48	752,6	903,1	1204,2	37,630	30,104	32	86	40	56
550	660	880	27,500	22	32	56	28	44	757,6	909,1	1212,2	37,880	30,304	24	100	44	40
552,8	663,4	884,5	27,640	22,112	40	86	56	72	762	914,4	1219,2	38,100	30,480	24	64	28	40
555,6	666,7	889	27,780	22,224	32	100	72	64	764	916,8	1222,4	38,200	30,560	24	100	48	44
558,4	670,1	893,4	27,920	22,336	28	86	44	40	767,8	921,4	1228,5	38,390	30,712	40	100	56	86
560	672	896	28	22,400	64	100	48	86	771,4	925,7	1234,2	38,570	30,856	28	72	32	48
562,6	675,1	900,2	28,130	22,504	40	100	64	72	777,8	933,4	1244,5	38,890	31,112	24	56	24	40
565,8	679	905,3	28,290	22,632	28	72	40	44	779,2	935	1246,7	38,960	31,168	28	100	44	48
568,2	681,8	909,1	28,410	22,728	32	100	44	40	785,8	943	1257,3	39,290	31,432	28	100	40	44
581,8	696,2	930,9	29,090	23,272	24	64	44	48	788,4	946,1	1261,4	39,420	31,536	24	86	40	44
584,4	701,3	935	29,220	23,376	56	100	44	72	789,8	947,8	1263,7	39,490	31,592	28	86	56	72
586,4	703,7	938,2	29,320	23,456	48	86	44	72	795,4	954,5	1272,6	39,770	31,816	32	100	44	56
587,8	705,4	940,5	29,390	23,512	28	72	56	64	800	960	1280	40	32	24	72	48	64
591,2	709,4	945,9	29,560	23,648	32	86	40	44	803,6	964,3	1285,8	40,180	32,144	32	100	56	72
595,2	714,2	952,3	29,760	23,808	28	100	48	40	806,2	967,4	1289,9	40,310	32,248	32	86	48	72
598	717,6	956,8	29,900	23,920	28	100	86	72	816,4	979,7	1306,2	40,820	32,656	28	100	56	64
600	720	960	30	24	28	56	32	48	818,2	981,8	1309,1	40,910	32,728	40	100	44	72
604,6	725,5	967,4	30,230	24,184	32	86	64	72	822,8	987,4	1316,5	41,140	32,912	28	72	40	64
606	727,2	969,6	30,300	24,240	48	100	44	64	825	990	1320	41,250	33	24	72	32	44
610,8	733	977,3	30,540	24,432	44	100	64	86	833,4	1000,1	1333,4	41,670	33,336	32	100	48	64
612,2	734,6	979,5	30,610	24,488	28	100	56	48	836,2	1003,4	1337,9	41,810	33,448	24	86	48	56
614,2	737	982,7	30,710	24,568	24	86	56	48	838,2	1005,8	1341,1	41,910	33,528	24	64	28	44
620,2	744,2	992,3	31,010	24,808	24	100	86	64	840	1008	1344	42	33,600	24	72	40	56
622,2	746,6	995,5	31,110	24,888	24	64	48	56	844,6	1013,5	1351,4	42,230	33,784	28	86	32	44
625	750	1000	31,250	25	28	100	64	56	855,6	1026,7	1369	42,780	34,224	24	56	24	44
628,6	754,3	1005,8	31,430	25,144	28	64	32	44	860	1032	1376	43	34,400	32	86	40	64
630	756	1008	31,500	25,200	32	72	40	56	872,8	1047,4	1396,5	43,640	34,912	24	72	44	64
635	762	1016	31,750	25,400	72	100	28	64	875	1050	1400	43,750	35	32	100	40	56
636,4	763,7	1018,2	31,820	25,456	44	100	40	56	879,6	1055,5	1407,4	43,980	35,184	32	86	44	72
640	768	1024	32	25,600	28	64	40	56	888,8	1066,6	1422,1	44,440	35,552	24	64	24	40
641,8	770,2	1026,9	32,090	25,672	24	56	32	44	892,8	1071,4	1428,5	44,640	35,712	28	100	32	40
645	774	1032	32,250	25,800	48	86	40	72	895,8	1075	1433,3	44,790	35,832	40	100	48	86
648,2	777,8	1037,1	32,410	25,928	24	100	72	56	900	1080	1440	45	36	28	72	32	56
651,6	781,9	1042,6	32,580	26,064	24	86	44	40	909	1090,8	1454,4	45,450	36,360	32	100	44	64
654,6	785,5	1047,4	32,730	26,184	32	72	44	64	912,2	1094,6	1459,5	45,810	36,488	24	86	44	56
660	792	1056	33	26,400	24	72	40	44	918,4	1102,1	1469,4	45,920	36,736	28	100	56	72
666,6	799,9	1066,6	33,330	26,664	24	100	40	32	921,4	1105,7	1474,2	46,070	36,856	28	86	48	72
670,2	804,2	1072,3	33,510	26,808	28	86	44	48	933,4	1120,1	1493,4	46,670	37,336	24	64	32	56
675,8	811	1081,3	33,790	27,032	28	86	40	44	937,6	1125,1	1500,2	46,880	37,504	32	100	48	72
678,8	814,6	1086,1	33,940	27,152	24	64	44	56	943	1131,6	1508,8	47,150	37,720	24	72	28	44

UNIVERSAL DIVIDING HEAD

TABLE FOR SPIRAL MILLING

HELIX LEAD					CHANGE GEARS				HELIX LEAD					CHANGE GEARS			
MM			INCHES		A	B	C	D	MM			INCHES		A	B	C	D
5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm					5 mm	6 mm	8 mm	4 TPI 6,35 mm	5 TPI 5,08 mm				
952,4	1142,9	1523,8	47,620	38,096	28	100	48	64	1612,5	1935	2580	80,625	64,500	32	86	28	84
955,6	1146,7	1529	47,780	38,224	24	86	48	64	1680	2016	2688	84	67,200	40	96	24	84
960	1152	1536	48	38,400	24	72	40	64	1720	2064	2732	86	68,800	40	96	24	86
967,6	1161,1	1548,2	48,380	38,704	32	86	40	72	1750	2100	2800	87,500	70	40	100	24	84
972,2	1166,6	1555,5	48,610	38,888	24	100	48	56	1777	2133	2844	88,850	71,080	24	80	24	64
977,2	1172,6	1563,5	48,860	39,088	40	100	44	86	1791	2150	2866	89,550	71,640	40	100	24	86
982,2	1178,6	1571,5	49,110	39,288	28	100	32	44	1800	2160	2880	90	72	28	84	24	72
985,4	1182,5	1576,6	49,270	39,416	24	86	32	44	1828	2194	2925	91,400	73,120	28	96	24	64
995,4	1194,5	1592,6	49,770	39,816	24	100	72	86	1842	2211	2948	92,100	73,680	32	96	28	86
1000	1200	1600	50	40	28	100	40	56	1866	2240	2986	93,300	74,640	24	84	24	64
1005,2	1206,2	1608,3	50,260	40,208	28	86	44	72	1875	2250	3000	93,750	75	32	100	28	84
1022,8	1227,4	1636,5	51,140	40,912	32	100	44	72	1904	2285	3047	95,200	76,160	28	100	24	64
1028,6	1234,3	1645,8	51,430	41,144	28	72	32	64	1919	2303	3071	95,950	76,760	32	100	28	86
1042,4	1250,9	1667,8	52,120	41,696	24	86	44	64	1944	2333	3111	97,200	77,760	24	100	24	56
1050	1260	1680	52,500	42	24	72	32	56	2000	2400	3200	100	80	28	84	24	80
1060,6	1272,7	1697	53,030	42,424	24	100	44	56	2047	2457	3276	102,350	81,880	28	86	24	80
1066,6	1279,9	1706,6	53,330	42,664	24	64	28	56	2057	2468	3291	102,850	82,280	28	96	24	72
1075	1290	1720	53,750	43	24	86	32	48	2083	2500	3333	104,150	83,320	32	100	24	80
1100	1320	1760	55	44	24	72	24	44	2100	2520	3360	105	84	24	84	24	72
1105,6	1326,7	1769	55,280	44,224	28	86	40	72	2133	2560	3413	106,650	85,320	24	96	24	64
1111,2	1333,4	1777,9	55,560	44,448	24	100	24	32	2142	2571	3428	107,100	85,680	32	100	28	96
1119,8	1343,8	1791,7	55,990	44,792	24	100	64	86	2150	2580	3440	107,500	86	28	86	24	84
1125	1350	1800	56,250	45	32	100	40	72	2187	2625	3500	109,350	87,480	32	100	24	84
1142,8	1371,4	1828,5	57,140	45,712	28	100	40	64	2222	2666	3555	111,100	88,880	24	100	24	64
1146	1375,2	1833,6	57,300	45,840	24	100	32	44	2239	2687	3583	111,950	89,560	32	100	24	86
1166,6	1399,9	1866,6	58,330	46,664	24	100	40	56	2285	2742	3657	114,250	91,400	28	96	24	80
1172,8	1407,4	1876,5	58,640	46,912	24	86	44	72	2333	2800	3733	116,650	93,320	24	84	24	80
1190,6	1428,7	1905	59,530	47,624	24	100	28	40	2380	2857	3809	119	95,200	28	100	24	80
1200	1440	1920	60	48	24	72	32	64	2388	2866	3822	119,400	95,520	24	86	24	80
1225	1470	1960	61,250	49	32	84	24	56	2400	2880	3840	120	96	28	96	24	84
1250	1500	2000	62,500	50	40	100	32	80	2457	2948	3931	122,850	98,280	28	96	24	86
1280	1536	2048	64	51,200	40	96	24	64	2500	3000	4000	125	100	28	100	24	84
1290	1548	2064	64,500	51,600	40	86	28	84	2508	3010	4013	125,400	100,320	24	96	24	84
1312,5	1575	2100	65,625	52,500	40	100	32	84	2559	3071	4095	127,950	102,360	28	100	24	86
1350	1620	2160	67,500	54	32	84	28	72	2666	3200	4266	133,300	106,640	24	96	24	80
1371,43	1645,71	2194,29	68,571	54,857	40	96	28	80	2800	3360	4480	140	112	24	96	24	84
1400	1680	2240	70	56	40	84	24	80	2857	3428	4571	142,850	114,280	28	100	24	90
1440	1728	2304	72	57,600	40	96	28	84	2866	3440	4586	143,300	114,640	24	96	24	80
1500	1800	2400	75	60	40	100	28	84	2916	3500	4666	145,800	116,640	24	100	24	84
1575	1890	2520	78,750	63	32	84	24	72	2986	3583	4777	149,300	119,440	24	100	24	86
1600	1920	2560	80	64	40	96	24	80	3333	4000	5333	166,650	133,320	24	100	24	96

NOTES. Spiral milling loose fixing screw 15 and introduces needle 42 in plate 4, so that movement may be driven to the main spindle.

For spiral leads greater than 50 mm the feed of the table must be automatically produced.

For spiral leads less than 50 mm, the automatic movement must be aid by the handle 13.

When the desired lead is too much short, driving may be direct between the table and the main spindle of the dividing head, disengaging clamp 37 to avoid any surplus effort in the gears.

Calculation will be

$$R = \frac{\text{Lead of the lead screw}}{\text{Lead of helix}} = \frac{\text{Driving gears}}{\text{Driven gears}}$$

Example: Lead of 4 mm in a milling machine with lead screw of 5 mm.

$$R = \frac{5}{4} = \frac{40}{32} \text{ gearing these wheels with idlers.}$$

